

MapAgent：面向地理空间推理的 动态地图工具集成分层智能体

中文对照翻译版

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MapAgent: A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration

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Agentic AI has significantly extended the capabilities of large language models (LLMs) by enabling complex reasoning and tool use. However, most existing frameworks are tailored to domains such as mathematics, coding, or web automation, and fall short on geospatial tasks that require spatial reasoning, multi-hop planning, and real-time map interaction. To address these challenges, we introduce **MapAgent**, a hierarchical multi-agent plug-and-play framework with customized toolsets and agentic scaffolds for map-integrated geospatial reasoning. Unlike existing flat agent-based approaches that treat tools uniformly—often overwhelming the LLM when handling similar but subtly different geospatial APIs—MapAgent decouples planning from execution. A high-level planner decomposes complex queries into subgoals, which are routed to specialized modules. For tool-heavy modules—such as map-based services—we then design a dedicated map-tool agent that efficiently orchestrates related APIs adaptively in parallel to effectively fetch geospatial data relevant for the query, while simpler modules (e.g., solution generation or answer extraction) operate without additional agent overhead. This hierarchical design reduces cognitive load, improves tool selection accuracy, and enables precise coordination across similar APIs. We evaluate MapAgent on four diverse geospatial benchmarks—MapEval-Textual, MapEval-API, MapEval-Visual, and MapQA—and demonstrate substantial gains over state-of-the-art tool-augmented and agentic baselines. We open-source our framework at [anonymous repository](#).

1 Introduction

Agentic large language models (LLMs) have significantly expanded the scope of AI systems by enabling complex reasoning, subgoal decomposition, and dynamic tool use (Yao et al., 2023; Kim et al., 2024; Du et al., 2024). Despite these ad-

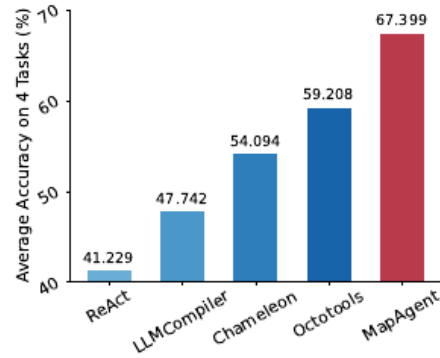


Figure 1: Performance comparison across 4 geo-spatial benchmarks. MapAgent significantly outperforms others and achieves state-of-the-art (SOTA) performance.

vances, most existing frameworks predominantly focus on domains such as mathematics (Lu et al., 2023, 2025), software engineering (Jimenez et al., 2023), and web automation (Qin et al., 2023; Du et al., 2024), while their application to **geospatial reasoning**—a ubiquitous capability that enables the automation of everyday tasks through natural language instructions, particularly in scenarios requiring interaction with specialized tools like map services—remains limited. This includes tasks such as route planning, location-based decision-making, spatial comparison, and finding nearby points of interest (e.g., restaurants, gas stations, or EV charging points) that satisfy user constraints.

Geospatial tasks pose distinct challenges compared to other reasoning problems. Queries in this domain often require multi-hop planning, dynamic spatial grounding, and coordination across multiple external APIs with overlapping but subtly different capabilities. For instance, solving queries like “Find the shortest route from home to the office with a stop at a highly rated coffee shop” or “Plan a three-day road trip from San Francisco to Yosemite

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Findings of the Association for Computational Linguistics: EACL 2026, 第 1296–1322 页

2026 年 3 月 24-29 日 ©2026 Association for Computational Linguistics

MapAgent：一种用于地理空间推理的分层智能体，具有动态

地图工具集成

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智能体式 AI (Agentic AI) 通过支持复杂推理和工具使用，显著扩展了大语言模型 (LLM) 的能力。然而，大多数现有框架是针对数学、编程或网页自动化等领域定制的，在需要空间推理、多跳规划和实时地图交互的地理空间任务上表现不足。为应对这些挑战，我们提出了 MapAgent，这是一种分层多智能体即插即用框架，具有定制化工具集和智能体式脚手架，用于地图集成的地理空间推理。与现有扁平化基于智能体的方法不同——这些方法统一对待工具，在处理相似但细微不同的地理空间 API 时常常使 LLM 不堪重负——MapAgent 将规划与执行解耦。高层规划器将复杂查询分解为子目标，并将其路由到专门模块。对于工具密集型模块——例如基于地图的服务——我们随后设计了一个专用地图工具智能体，它能够自适应地并行高效编排相关 API，以有效获取与查询相关的地理空间数据，而更简单的模块（例如解决方案生成或答案提取）则无需额外的智能体开销即可运行。这种分层设计降低了认知负荷，提高了工具选择准确性，并实现了相似 API 之间的精确协调。我们在四个多样化的地理空间基准上评估 MapAgent——MapEval-Textual、MapEval-API、MapEval-Visual 和 MapQA——并展示了相对于最先进的工具增强和智能体式基线方法的显著提升。我们在匿名仓库中开源了我们的框架。

1 引言

智能体式大语言模型 (LLM) 通过支持复杂推理、子目标分解和动态工具使用，显著扩展了 AI 系统的范围 (Yao et al., 2023; Kim et al., 2024; Du et al., 2024)。尽管取得了这些进

ReAct

LLMCompilerChameleonOctotoolsMapAgent

40

50

60

70

4 个任务上的平均准确率 (%)

41.229

47.742

54.094

59.208

67.399

图 1：4 个地理空间基准上的性能比较。MapAgent 显著优于其他方法，并达到了最先进 (SOTA) 性能。

展，大多数现有框架主要关注数学 (Lu et al., 2023, 2025)、软件工程 (Jimenez et al., 2023) 和网页自动化 (Qin et al., 2023; Du et al., 2024) 等领域，而它们在地理空间推理中的应用——这是一种无处不在的能力，可通过自然语言指令实现日常任务自动化，尤其是在需要与地图服务等专用工具交互的场景中——仍然有限。这包括路线规划、基于位置的决策、空间比较，以及寻找满足用户约束的附近兴趣点（例如餐厅、加油站或 EV 充电点）等任务。

与其他推理问题相比，地理空间任务提出了独特的挑战。该领域中的查询通常需要多跳规划、动态空间

grounding，以及跨多个外部 API 的协调，这些 API 具有重叠但细微不同的能力。例如，解决诸如“找到从家到办公室、并在高评分咖啡店停留的最短路线”或“规划一次从旧金山到优胜美地的三日公路旅行”

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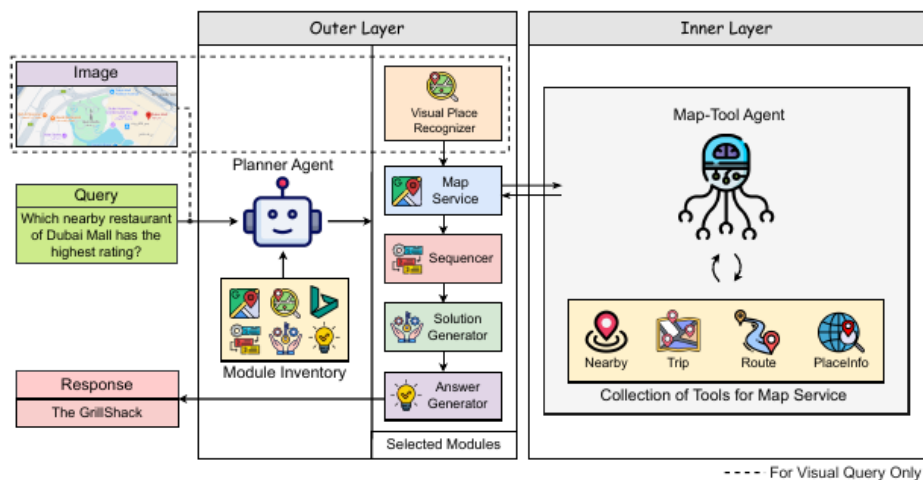


Figure 2: Overview of MapAgent. Given a user query (optionally with an image), the Planner Agent decomposes it into subtasks using the available Module Inventory and selects appropriate modules for each subtask. For tool-heavy modules, a dedicated agent (e.g., a Map-Tool Agent) manage interactions with the associated tools adaptively.

with overnight stops at scenic towns and hiking spots along the way” involves chaining services for nearby place search, distance estimation, POI (point-of-interest) retrieval, temporal scheduling, and real-time filtering. Despite addressing very related geospatial functions, the available Map APIs vary in their own schema, constraints, and precious functionalities, making it non-trivial to integrate them effectively. The agent must not only orchestrate multiple tools, but also combine and reason over the retrieved data to produce timely, coherent responses that preserve geospatial consistency. However, current agentic systems are not designed for this setting. Most plug-and-play tool-using agents adopt flat execution architectures that treat tools as generic, interchangeable black boxes. This leads to two major limitations in the context of geospatial reasoning: (i) *Tool inflation*: Existing systems often bundle disparate tools—one of each kind—without accounting for the fine-grained functional variations present within map services. As geospatial APIs proliferate (e.g., distance, proximity, directions, routing, nearby search, place details, timelines), agents face a combinatorial burden in both planning and execution, increasing decision complexity and reducing overall effectiveness.

(ii) *Tool incapability*: Tools integrated into existing agents are typically primitive (single-API-based) and generic (e.g., image captioning, table lookup, code generation, or web search), and are

not designed for the rich, mixed-mode (parallel and sequential) interactions required by real-world map services like Google Maps. While some API calls—such as computing distances and retrieving place details—can be performed in parallel, many tasks require tightly coordinated sequential steps, such as fetching detailed information only for places identified in a prior nearby search.

To address these challenges, we propose **MapAgent**, a hierarchical plug-and-play multi-agent framework designed for map-integrated geospatial reasoning. First, to overcome the issue of *tool incapability*, we design a set of four heterogeneous map tools, each composed of one or more primitive APIs tailored to perform key geospatial operations. These tools encapsulate common spatial functionalities—such as nearby search, route planning, and place detail retrieval—and are constructed to support both parallel and sequential execution flows. By abstracting low-level primitive API calls into higher-level tool interfaces, MapAgent enables robust handling of mixed-mode reasoning: for example, retrieving details of candidate places found via a nearby search in a sequential pipeline, while simultaneously computing alternative routes in parallel. This structured orchestration supports the complex, map-centric reasoning patterns required for real-world spatial tasks.

Second, to mitigate the problem of *tool inflation*, MapAgent adopts a hierarchical architecture

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迪拜购物中心附近哪家餐厅评分最高？
查询
响应
The GrillShack
解决方案生成器
答案生成器
视觉地点识别器
地图服务
外层 内层
图像

选定模块
仅用于视觉查询
序列化器
地图工具智能体 (Map-Tool Agent)
附近行程
路线
地点信息
地图服务工具集合
规划智能体 (Planner Agent)
模块清单

图2：MapAgent概览。给定用户查询（可选附带图像），规划智能体（Planner Agent）利用可用的模块清单将其分解为子任务，并为每个子任务选择合适的模块。对于工具密集型模块，由专用智能体（例如地图工具智能体（Map-Tool Agent））自适应地管理与相关工具的交互。

“沿途在风景优美的城镇和徒步点过夜”涉及串联以下服务：附近地点搜索、距离估计、POI（兴趣点）检索、时间调度和实时过滤。尽管这些服务处理的是高度相关的地理空间功能，但可用的地图API在各自的模式、约束和宝贵功能上各不相同，使得有效集成它们并非易事。智能体不仅必须编排多个工具，还必须对检索到的数据进行组合和推理，以产生及时、连贯且保持地理空间一致性的响应。

然而，当前的智能体系统并非为此场景而设计。大多数即插即用的工具使用型智能体采用扁平执行架构，将工具视为通用、可互换的黑箱。这在地理空间推理背景下导致两个主要局限：(i) 工具膨胀：现有系统通常捆绑互不相同的工具——每种各一个——而未考虑地图服务中存在的细粒度功能差异。随着地理空间API的激增（例如，距离、邻近度、方向、路线规划、附近搜索、地点详情、时间线），智能体在规划和执行中面临组合负担，增加了决策复杂度并降低了整体有效性。(ii) 工具能力不足：集成到现有智能体中的工具通常是原始的（基于单一API）和通用的（例如，图像描述、表格查找、代码生成或网络搜索），并非为Google Maps等真实世界地图服务所需的丰富混合模式（并行和顺序）交互而设计。虽然某些API调用——例如计算距离和检索地点详情——可以并行执行，但许多任务需要紧密协调的顺序步骤，例如仅对先前附近搜索中识别出的地点获取详细信息。

为应对这些挑战，我们提出MapAgent，一个为地图集成地理空间推理设计的分层即插即用多智能体框架。首先，为克服工具能力不足的问题，我们设计了一组四个异构地图工具，每个工具由一个或多个原始API组成，专门用于执行关键地理空间操作。这些工具封装了常见的空间功能——例如附近搜索、路线规划和地点详情检索——并构建为支持并行和顺序执行流程。通过将底层原始API调用抽象为更高层的工具接口，MapAgent能够稳健地处理混合模式推理：例如，在顺序流水线中检索通过附近搜索找到的候选地点的详情，同时并行计算备选路线。这种结构化编排支持真实世界空间任务所需的复杂、以地图为中心的推理模式。

其次，为缓解工具膨胀问题，MapAgent采用分层架构

Method	Parallel Tool Calling	Compositional Reasoning	Hierarchical	Plug-n-play or Map-tools
ReAct (Yao et al., 2023)	✗	✗	✗	✗
LLMCompiler (Kim et al., 2024)	✓	✗	✗	✗
Octotools (Lu et al., 2025)	✓	✓	✗	✓
Chameleon (Lu et al., 2023)	✗	✓	✗	✓
Anytool (Du et al., 2024)	✗	✗	✓	✗
MapAgent (Ours)	✓	✓	✓	✓

Table 1: Feature comparison of different tool calling frameworks in context to map-integrated geospatial reasoning.

that decouples high-level task planning from low-level tool execution. A top-level planner agent decomposes complex natural language queries into structured subgoals and routes them to appropriate tool workflows. For tool-intensive subgoals—such as those involving map services—we introduce a dedicated *map-tool agent* that adaptively manages interactions with multiple map tools, issuing parallel API calls where appropriate and coordinating sequential workflows when required. In contrast, lightweight tasks—such as answer formatting or preference synthesis—are handled directly without additional agent overhead.

We evaluate MapAgent across four diverse geospatial reasoning benchmarks—MapQA, MapEval-Textual, MapEval-Visual, and MapEval-API—each presenting distinct challenges, including textual context, multimodal inputs, and query-only settings. Built using both open-weight and closed-source backbone LLMs, MapAgent consistently outperforms prior state-of-the-art agentic and tool-augmented LLM frameworks across all benchmarks. Utilizing GPT-3.5-Turbo, it achieves a 10% improvement on both the MapEval-API and MapEval-Textual datasets, and an 11.22% improvement on the MapQA dataset over the strong baseline OctoTools (Lu et al., 2025). Furthermore, with GPT-4o, MapAgent achieves a 4.41% improvement on the MapEval-Visual dataset compared with the same baseline. Overall, MapAgent improves performance by 8.2% on average over the strongest baseline, OctoTools, in geospatial tasks that require multi-hop planning, spatial inference, and dynamic tool use. Our ablations, qualitative, and fine-grained analyses further provide more insights into the effectiveness of MapAgent.

2 Related Work

Compositional Reasoning with LLMs. Solving complex multi-step problems often involves

decomposing tasks into manageable sub-problems (Perez et al., 2020; Khot et al., 2023). Prompt-based methods like Chain-of-Thought (Wei et al., 2022), Least-to-Most (Zhou et al., 2023), ReAct (Yao et al., 2023), Pearl (Sun et al., 2024), Forest-of-Thought (Bi et al., 2024), and rStar-Math (Guan et al., 2025) have advanced the ability of LLMs to reason sequentially. Other approaches synthesize structured programs from natural language (e.g., LLM Compiler (Kim et al., 2024)) or orchestrate modular tools (e.g., Chameleon (Lu et al., 2023), OctoTools (Lu et al., 2025)). While effective, these systems often assume flat or static module inventories and struggle with tool variants that differ subtly in schema or functionality. In contrast, MapAgent leverages hierarchical planning and dynamic module selection to support plug-and-play coordination over overlapping but heterogeneous APIs, particularly in the geospatial domain.

Geospatial Reasoning with Language Models. Geospatial QA has been studied through rule-based and template-driven systems that convert natural language into structured queries over static databases (e.g., PostGIS, DBpedia (Auer et al., 2007), YAGO (Suchanek et al., 2007), Freebase (Bollacker et al., 2007), OSM). Notable examples include GeoSPARQL-based systems (Car et al., 2022) and datasets such as GeoQuestions1089 (Kefalidis et al., 2023) and YAGO2geo (Karalis et al., 2019). These methods are precise but inflexible, limited by their reliance on static schemas and query grammars.

Recent work explores LLMs’ intrinsic geographic knowledge (e.g., GPT4GEO (Roberts et al., 2023), (Mai et al., 2024)) or their capacity to answer map-related queries directly (e.g., MapQA (Li et al., 2025), MapEval (Dihan et al., 2025b,a)). While promising, these models lack mechanisms to integrate external tools or coordinate multi-API workflows. MapAgent builds on this line of

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方法 并行 组合式 分层 即插即用

工具调用 推理 或 Map-tools

ReAct (Yao et al., 2023) $\times \times \times \times$

LLMCompiler (Kim et al., 2024) $\checkmark \times \times \times$

Octotools (Lu et al., 2025) $\checkmark \checkmark \times \checkmark$

Chameleon (Lu et al., 2023) $\times \checkmark \times \checkmark$

Anytool (Du et al., 2024) $\times \times \checkmark \times$

MapAgent (Ours) $\checkmark \checkmark \checkmark \checkmark$

表 1：在面向地图集成地理空间推理的背景下，不同工具调用框架的功能比较。

其将高层任务规划与底层工具执行解耦。一个顶层规划器智能体 (planner agent) 将复杂的自然语言查询分解为结构化子目标，并将它们路由到适当的工具工作流。对于工具密集型子目标——例如涉及地图服务的子目标——我们引入一个专用的地图工具智能体 (map-tool agent)，它自适应地管理与多个地图工具的交互，在适当的时候发起并行 API 调用，并在需要时协调顺序工作流。相比之下，轻量级任务——例如答案格式化或偏好综合——则直接处理，不增加额外的智能体开销。

我们在四个多样化的地理空间推理基准上评估 MapAgent——MapQA、MapEval-Textual、MapEval-Visual 和 MapEval-API——每个基准都呈现出不同的挑战，包括文本上下文、多模态输入和仅查询设置。MapAgent 使用开放权重和闭源骨干 LLM 构建，在所有基准上始终优于此前的 state-of-the-art 智能体式和工具增强型 LLM 框架。使用 GPT-3.5-Turbo 时，它在 MapEval-API 和 MapEval-Textual 数据集上均实现了 10% 的提升，并在 MapQA 数据集上相较于强基线 OctoTools (Lu et al., 2025) 实现了 11.22% 的提升。此外，在使用 GPT-4o 时，MapAgent 在 MapEval-Visual 数据集上相较于同一基线实现了 4.41% 的提升。总体而言，在需要多跳规划、空间推理和动态工具使用的地理空间任务中，MapAgent 相较于最强基线 OctoTools 平均提升了 8.2%。我们的消融、定性和细粒度分析进一步为 MapAgent 的有效性提供了更多见解。

2 相关工作

使用 LLM 的组合式推理。解决复杂的多步问题通常涉及将任务分解为可管理的子问题 (Perez et al., 2020; Khot et al., 2023)。基于提示的方法，如 Chain-of-Thought (Wei et al., 2022)、Least-to-Most (Zhou et al., 2023)、ReAct (Yao et al., 2023)、Pearl (Sun et al., 2024)、Forest-of-Thought (Bi et al., 2024) 和 rStar-Math (Guan et al., 2025)，提升了 LLM 进行顺序推理的能力。其他方法从自然语言合成结构化程序 (例如 LLM Compiler (Kim et al., 2024)) 或编排模块化工具 (例如 Chameleon (Lu et al., 2023)、OctoTools (Lu et al., 2025))。尽管这些系统有效，但它们通常假设模块清单是扁平的或静态的，并且难以处理在模式或功能上存在细微差异的工具变体。相比之下，MapAgent 利用分层规划和动态模块选择，以支持在重叠但异构的 API 上进行即插即用协调，尤其是在地理空间领域。

使用语言模型进行地理空间推理。地理空间问答已通过基于规则和模板驱动的系统进行研究，这些系统将自然语言转换为对静态数据库的结构化查询 (例如 PostGIS、DBpedia (Auer et al., 2007)、YAGO (Suchanek et al., 2007)、Freebase (Bollacker et al., 2007)、OSM)。值得注意的例子包括基于 GeoSPARQL 的系统 (Car et al., 2022) 以及诸如 GeoQuestions1089 (Kefalidis et al., 2023) 和 YAGO2geo (Karalis et al., 2019) 等数据集。这些方法精确但缺乏灵活性，受限于其对静态模式和查询语法的依赖。

近期工作探索了 LLM 的内在地理知识 (例如 GPT4GEO (Roberts et al., 2023)、(Mai et al., 2024)) 或其直接回答地图相关查询的能力 (例如 MapQA (Li et al., 2025)、MapEval (Dihan et al., 2025b,a))。尽管这些模型前景可观，但它们缺乏整合外部工具或协调多 API 工作流的机制。MapAgent 建立在这一研究方向之上。

work by introducing a multi-agent, tool-augmented framework that dynamically coordinates geospatial modules across modalities—including textual, visual, and API inputs—offering enhanced reasoning capabilities for real-world map-based tasks.

3 Method

We propose **MapAgent**, a hierarchical plug-and-play multi-agent framework for map-integrated geospatial reasoning. MapAgent addresses two core limitations in current agentic systems—*tool incapability* and *tool inflation*—through a structured two-layer scaffold that cleanly separates reasoning, planning, and tool execution.

3.1 Overview

We introduce **MapAgent**, a hierarchical multi-agent framework for solving complex geospatial reasoning tasks via real-world map services. Real-world queries often require chaining spatial operations, querying external APIs, and integrating geographic evidence—capabilities that single-shot prompting lacks. To address this, MapAgent enables structured planning, map-tool composition, and spatial reasoning through specialized agents orchestrated in a hierarchical coordination.

In MapAgent, a top-level *planner agent* $\mathcal{P}$ first decomposes the user query x into subgoals, enabling modular execution and interpretable reasoning. These are handled by specialized *functional modules* M , which delegate tasks such as filtering or aggregation to dedicated components. Figure 1 shows an overview of MapAgent.

Geospatial subgoals are routed to a *Map-Tool Agent* $\mathcal{M}_{\text{map}}$, which composes four core tools Nearby, PlaceInfo, Route, and Trip—each built over real-world Google Map APIs to support spatial search, detail retrieval, and routing. This modular design allows MapAgent to solve complex, multi-hop spatial queries by dynamically planning over tool-based workflows. We describe the architecture in Section 3.2, the planner agents and modules in Section 3.3, the map tools in Section 3.4.

3.2 Scaffolded Architecture (Hierarchy)

MapAgent comprises two conceptual layers:

Planner Layer (Top-level): A planner agent decomposes the input x into a sequence of semantically coherent subgoals $[g_1, \dots, g_n]$, and routes

each g_i to a functional module or tool chain based on a structured inventory (discussed in Table 10).

Execution Layer (Bottom-level): This layer manages module-specific execution logic. It includes both lightweight modules (e.g., sequencing, formatting) and a specialized *Map-Service Module*, which integrates a dedicated **Map-Tool Agent** responsible for executing geospatial tasks using a curated set of map tools.

3.3 Module Inventory and Planner Agent

The planner agent $\mathcal{P}$ (based on an LLM or VLLM) receives the input x , an inventory M of available module to perform the subgoals, and outputs a structured execution plan $\pi = [(g_1, m_1), \dots, (g_n, m_n)]$, where each g_i is a subgoal assigned to a module $m_i \in M$. Module Inventory M contains several task-specific modules:

Visual Place Recognizer: Given an image map-snapshot, using the corresponding VLLM, this module extracts the central geographic place (name) in the image and estimates the geographic scope (approximate radius) covered in the image.

Sequencer: This module is responsible for organizing and structuring the responses received from preceding modules. It arranges unstructured information into a logical sequence using LLMs.

Solution Generator: This module synthesizes the final answer to the user query using the corresponding LLM or VLM. It takes structured information and generates a comprehensive response.

Answer Generator: This module focuses on refining and verifying the answer generated by the Solution Generator. It extracts the predicted answer and checks its consistency.

Map-Service Module: A tool-heavy module managed by the Map-Tool Agent to handle core Map-tool interactions.

3.4 Map-Service Module: Map-Tool Agent and Tool Design

Map-Tool Agent $\mathcal{M}_{\text{map}}$ takes as input the query x , along with an optional place name and radius inferred by the Visual Place Recognizer for multi-modal queries. It then adaptively interacts with a suite of geospatial tools designed to fetch the necessary spatio-temporal data from underlying map services (e.g., Google Maps).

We design four heterogeneous **MapTools**, each composed of one or more primitive APIs from Google Map (e.g., directions, place info) to support

通过引入一个多智能体、工具增强的框架，该框架能够跨模态（包括文本、视觉和 API 输入）动态协调地理空间模块，从而为现实世界中基于地图的任务提供增强的推理能力。

3 方法

我们提出 MapAgent，一个用于地图集成地理空间推理的分层即插即用多智能体框架。MapAgent 通过一个结构化的两层脚手架解决了当前智能体系统中的两个核心局限——工具能力不足（tool incapability）和工具膨胀（tool inflation），该脚手架清晰地将推理、规划和工具执行分离。

3.1 概述

我们介绍 MapAgent，一个分层多智能体框架，用于通过真实世界地图服务解决复杂的地理空间推理任务。真实世界的查询通常需要串联空间操作、查询外部 API 并整合地理证据——这些能力是单次提示（single-shot prompting）所缺乏的。为解决这一问题，MapAgent 通过分层协调编排的专用智能体，实现结构化规划、地图工具组合和空间推理。

在 MapAgent 中，顶层规划智能体 P 首先将用户查询 x 分解为子目标，从而实现模块化执行和可解释推理。这些子目标由专用功能模块 M 处理，它们将过滤或聚合等任务委托给专门的组件。图 1 展示了 MapAgent 的概览。地理空间子目标被路由到 Map-Tool Agent Mmap，它组合了四个核心工具——Nearby、PlaceInfo、Route 和 Trip——每个工具都构建在真实世界的 Google Map API 之上，以支持空间搜索、详情检索和路线规划。这种模块化设计使 MapAgent 能够通过基于工具的工作流上进行动态规划，解决复杂的多跳空间查询。我们在 3.2 节描述架构，3.3 节描述规划智能体和模块，3.4 节描述地图工具。

3.2 脚手架架构（层次结构）

MapAgent 包含两个概念层：

规划层（顶层）：规划智能体将输入 x 分解为一系列语义连贯的子目标 $[g_1, \dots, g_n]$ ，并根据结构化清单（在表 10 中讨论）将每个 g_i 路由到功能模块或工具链。

执行层（底层）：该层管理模块特定的执行逻辑。它包括轻量级模块（例如，排序、格式化）和一个专门的 Map-Service Module，后者集成了一个专用的 Map-Tool Agent，负责使用一组精选的地图工具执行地理空间任务。

3.3 模块清单与规划智能体

规划智能体 P（基于 LLM 或 VLLM）接收输入 x 、可用模块清单 M 以执行子目标，并输出结构化执行计划 $\pi = [(g_1, m_1), \dots, (g_n, m_n)]$ ，其中每个 g_i 是分配给模块 $m_i \in M$ 的子目标。模块清单 M 包含若干任务特定模块：

Visual Place Recognizer：给定一张图像地图快照，使用相应的 VLLM，该模块提取图像中的中心地理位置（名称）并估计图像覆盖的地理范围（近似半径）。

Sequencer：该模块负责组织和结构化来自前述模块的响应。它使用 LLM 将非结构化信息整理为逻辑序列。

Solution Generator：该模块使用相应的 LLM 或 VLM 综合出对用户查询的最终答案。它接收结构化信息并生成全面的响应。

Answer Generator：该模块专注于细化和验证由 Solution Generator 生成的答案。它提取预测答案并检查其一致性。

Map-Service Module：一个由 Map-Tool Agent 管理的工具密集型模块，用于处理核心的 Map-tool 交互。

3.4 Map-Service Module：Map-Tool Agent 与工具设计

Map-Tool Agent Mmap 接收查询 x ，以及对于多模态查询由 Visual Place Recognizer 推断的可选地点名称和半径。然后，它自适应地与一套地理空间工具交互，这些工具旨在从底层地图服务（例如，Google Maps）获取必要的时空数据。

我们设计了四个异构的 MapTools，每个工具由一个或多个来自 Google Map 的原始 API（例如，directions、place info）组成，以支持

LLM	MapEval-API						
	Method	Overall	Place Info	Nearby	Routing	Trip	Unanswerable
GPT-3.5-Turbo	ReAct	27.33	39.06	22.89	33.33	19.40	15.00
	LLMCompiler	44.75	53.13	51.80	48.48	37.10	0.00
	Chameleon	50.45	54.54	54.10	52.00	43.07	5.00
	Octotools	56.27	76.56	62.65	56.06	40.30	5.00
	MapAgent	66.31	88.24	77.05	68.18	49.23	20.00
Qwen-2.5-72B	ReAct	41.56	55.44	38.25	42.40	32.75	0.00
	LLMCompiler	49.63	59.25	55.29	51.23	40.77	0.00
	Chameleon	52.70	61.30	60.29	55.77	46.55	10.00
	Octotools	59.40	80.95	71.08	59.09	42.42	0.00
	MapAgent	70.94	89.40	82.85	69.76	54.54	25.00

Table 2: Accuracy (%) of different methods and backbones on the MapEval-API Dataset.

LLM	MapEval-Textual						
	Method	Overall	Place Info	Nearby	Routing	Trip	Unanswerable
GPT-3.5-Turbo	Base Model	37.67	26.56	53.01	48.48	28.36	5.00
	ReAct	42.25	42.90	48.18	50.67	35.34	0.00
	LLMCompiler	49.23	58.44	56.98	53.33	40.81	0.00
	Chameleon	56.83	60.77	62.50	57.20	47.38	5.00
	Octotools	62.78	84.22	68.92	61.67	44.33	5.00
	MapAgent	72.94	90.06	82.76	73.23	54.15	20.00
Qwen-2.5-72B	Base Model	57.00	62.50	71.08	63.64	41.79	10.00
	ReAct	58.44	64.44	68.25	56.80	46.28	5.00
	LLMCompiler	62.58	74.40	75.38	64.22	48.32	5.00
	Chameleon	65.75	76.40	72.54	70.60	52.26	10.00
	Octotools	70.25	85.55	77.22	67.20	53.30	10.00
	MapAgent	76.24	94.20	84.75	74.80	59.20	20.00

Table 3: Accuracy (%) of different methods and backbones on the MapEval-Textual Dataset.

essential geospatial operations:

Trip Tool: Retrieves information about locations and travel routes between them. Combines the Place Details and Directions APIs to fetch place metadata and step-by-step route instructions.

Route Tool: Fetches route information between two points using the Directions API, including distance, estimated time, and navigation steps.

Nearby Tool: Identifies points of interest within a specified area using the Nearby Search API. Returns place names, ratings, and other metadata.

PlaceInfo Tool: Retrieves detailed information about a given place using the Place Details API, including address, hours, contact info, and reviews.

4 Experimental Setup

4.1 Datasets

We evaluated MapAgent on four benchmarks spanning two modalities as shown in Table 4. These

benchmarks encompass a wide range of complex geospatial reasoning tasks, including long-context reasoning, API interactions, visual map analysis, numerical calculation, and multi-step reasoning. Three benchmarks are from MapEval (Dihan et al., 2025b):

Dataset	# Tasks	Modality
MapEval-Textual	300	Text
MapEval-API	300	Text
MapEval-Visual	400	Vision
MapQA	3154	Text

Table 4: Overview of datasets: statistics and modality

MapEval-Textual: Given a user query with a long textual context describing map locations, POIs, routes, navigation details, travel distances/times, and user-generated content such as ratings/reviews, the task is to answer the query based on the context.

MapEval-API: Given a user query and access to map-based APIs, the task is to answer the query

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LLM

MapEval-API

Method Overall Place Info Nearby Routing Trip Unanswerable

GPT-3.5-Turbo

ReAct	27.33	39.06	22.89	33.33	19.40	15.00
LLMCompiler	44.75	53.13	51.80	48.48	37.10	0.00
Chameleon	50.45	54.54	54.10	52.00	43.07	5.00
Octotools	56.27	76.56	62.65	56.06	40.30	5.00
MapAgent	66.31	88.24	77.05	68.18	49.23	20.00

Qwen-2.5-72B

ReAct	41.56	55.44	38.25	42.40	32.75	0.00
LLMCompiler	49.63	59.25	55.29	51.23	40.77	0.00
Chameleon	52.70	61.30	60.29	55.77	46.55	10.00
Octotools	59.40	80.95	71.08	59.09	42.42	0.00
MapAgent	70.94	89.40	82.85	69.76	54.54	25.00

表 2：不同方法和骨干模型在 MapEval-API 数据集上的准确率（%）。

LLM

MapEval-Textual

Method Overall Place Info Nearby Routing Trip Unanswerable

GPT-3.5-Turbo

Base Model	37.67	26.56	53.01	48.48	28.36	5.00
ReAct	42.25	42.90	48.18	50.67	35.34	0.00
LLMCompiler	49.23	58.44	56.98	53.33	40.81	0.00
Chameleon	56.83	60.77	62.50	57.20	47.38	5.00
Octotools	62.78	84.22	68.92	61.67	44.33	5.00
MapAgent	72.94	90.06	82.76	73.23	54.15	20.00

Qwen-2.5-72B

Base Model	57.00	62.50	71.08	63.64	41.79	10.00
ReAct	58.44	64.44	68.25	56.80	46.28	5.00
LLMCompiler	62.58	74.40	75.38	64.22	48.32	5.00

Chameleon	65.75	76.40	72.54	70.60	52.26	10.00
Octotools	70.25	85.55	77.22	67.20	53.30	10.00
MapAgent	76.24	94.20	84.75	74.80	59.20	20.00

表 3：不同方法和骨干模型在 MapEval-Textual 数据集上的准确率（%）。

essential geospatial operations:

Trip Tool：检索位置信息
以及它们之间的旅行路线。结合了
Place Details 和 Directions API，以获取地点
元数据和逐步路线指引。

Route Tool：使用 Directions API 获取两点之间的路线信息，
包括距离、预计时间和导航步骤。

Nearby Tool：使用 Nearby Search API 识别指定区域内的兴趣点。

返回地点名称、评分和其他元数据。
PlaceInfo Tool：使用 Place Details API 检索给定地点的详细信息，包括地址、营业时间、联系信息和评论。

4 实验设置

4.1 数据集

我们在四个基准上评估了 MapAgent，涵盖两种模态，如表 4 所示。这些基准涵盖了广泛的复杂地理空间推理任务，包括长上下文推理、API 交互、视觉地图分析、数值计算和多步推理。其中三个基准来自 MapEval (Dihan et al., 2025b)：

Dataset #	Tasks	Modality
MapEval-Textual	300	Text
MapEval-API	300	Text
MapEval-Visual	400	Vision
MapQA	3154	Text

表 4：数据集概览：统计和模态

MapEval-Textual：给定一个用户查询，并附带描述地图位置、POI、路线、导航细节、旅行距离/时间和用户生成内容（如评分/评论）的长文本上下文，任务是基于该上下文回答查询。
MapEval-API：给定一个用户查询并可访问基于地图的 API，任务是回答查询
1300

VLM	MapEval-Visual						
	Method	Overall	Place Info	Nearby	Routing	Counting	Unanswerable
GPT-4o	Base Model	58.90	76.86	57.78	50.00	47.73	40.00
	ReAct	58.50	78.28	58.24	52.29	53.11	0.00
	LLMCompiler	61.88	81.29	60.80	54.42	55.22	0.00
	Chameleon	62.51	82.44	62.77	55.67	57.18	5.00
	Octotools	64.54	84.20	64.53	53.44	61.36	5.00
	MapAgent	68.95	88.64	63.54	62.50	63.63	25.00
Qwen-2.5-VL-72B	Base Model	60.35	76.86	54.44	43.04	52.33	60.00
	ReAct	59.18	74.54	54.40	56.40	55.40	10.00
	LLMCompiler	64.28	82.50	61.80	58.88	59.80	10.00
	Chameleon	66.40	83.46	64.32	59.30	62.35	15.00
	Octotools	69.50	88.78	66.48	63.22	64.28	20.00
	MapAgent	72.30	89.45	66.25	67.50	66.50	40.00

Table 5: Accuracy (%) of different methods and backbones on the MapEval-Visual Dataset.

based on the structured data retrieved via the APIs. **MapEval-Visual:** Given a user query and a digital map snapshot showing spatial layouts, routes, landmarks, OCR text (e.g., ratings), and symbolic elements (e.g., icons or signs), the task is to answer the query based on the visual context. **MapQA (Li et al., 2025):** A geospatial QA benchmark constructed from OpenStreetMap using SQL query templates. It includes QA pairs covering nine reasoning types (e.g., neighborhood inference, spatial proximity, type classification), grounded in real-world geo-entity geometries.

4.2 Baseline and Metric

MapAgent is a multi-agent compositional reasoning framework. We therefore compare its performance against agentic frameworks, such as ReAct (Yao et al., 2023) and LLMCompiler (Kim et al., 2024), and compositional reasoning frameworks, such as Chameleon (Lu et al., 2023) and OctoTool (Lu et al., 2025). For the MapEval-API, MapEval-Textual and MapQA datasets, we use GPT-3.5-Turbo (?) and Qwen-2.5-72b (?). For the MapEval-Visual dataset, we use models served via vLLM, specifically GPT-4o (?) and Qwen-2.5-VL-72b (?). We evaluate model performance using accuracy, expressed as a percentage.

4.3 Results

The evaluation of the MapAgent framework across four distinct and challenging map-related datasets reveals its significant advancements in accuracy, consistently outperforming existing agentic approaches and demonstrating robust performance irrespective of the underlying language model back-

bone. These findings underscore MapAgent’s effectiveness as a robust solution for a broad spectrum of map-based query processing tasks.

Strong Gains Across Benchmarks Across all evaluated datasets, the MapAgent framework consistently achieves high accuracy. On the MapEval-API dataset in Table 2, MapAgent achieves overall accuracies of roughly 70% and 72% when powered by GPT-3.5-Turbo and Qwen-2.5-72B, respectively. Similarly, on the MapEval-Textual dataset in Table 3, MapAgent attains accuracies of 72.94% with GPT-3.5-Turbo and 76.24% with Qwen-2.5-72B. The MapEval-Visual results in Table 5 further reinforce this trend, with MapAgent achieving the highest overall accuracy—68.95% using GPT-4o and 72.30% with Qwen-2.5-VL-72B. Finally, on the MapQA benchmark in Table 6, MapAgent achieves the highest overall accuracy of 55.58% with GPT-3.5-Turbo and 55.93% with Qwen-2.5-72B.

Surpassing Prior Methods The performance of MapAgent consistently surpasses that of other prominent methods, including OctoTools, Chameleon, ReAct, and LLMCompiler—notably the second-best performing OctoTools—across all evaluated datasets and language models. In Figure 5, on the MapEval-API dataset, MapAgent shows accuracy gains of approximately 10% and 11.54% over OctoTools when using GPT-3.5-Turbo and Qwen-2.5-72B, respectively. For the MapQA benchmark, MapAgent outperforms OctoTools by approximately 11.22% with GPT-3.5-Turbo and 9% with Qwen-2.5-72B. Lastly, on the MapEval-Textual dataset, MapAgent’s accuracy

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VLM

MapEval-Visual

方法 总体 地点 信息 附近 路线规划 计数 无法回答

GPT-4o

Base Model 58.90 76.86 57.78 50.00 47.73 40.00

ReAct 58.50 78.28 58.24 52.29 53.11 0.00

LLMCompiler 61.88 81.29 60.80 54.42 55.22 0.00

Chameleon 62.51 82.44 62.77 55.67 57.18 5.00

Octotools 64.54 84.20 64.53 53.44 61.36 5.00

MapAgent 68.95 88.64 63.54 62.50 63.63 25.00

Qwen-2.5-VL-72B

Base Model 60.35 76.86 54.44 43.04 52.33 60.00

ReAct 59.18 74.54 54.40 56.40 55.40 10.00

LLMCompiler 64.28 82.50 61.80 58.88 59.80 10.00

Chameleon 66.40 83.46 64.32 59.30 62.35 15.00

Octotools 69.50 88.78 66.48 63.22 64.28 20.00

MapAgent 72.30 89.45 66.25 67.50 66.50 40.00

表 5：不同方法和骨干模型在 MapEval-Textual 数据集上的准确率（%）。

基于通过 API 检索到的结构化数据。

MapEval-Visual：给定一个用户查询和一张数字地图快照，其中展示了空间布局、路线、地标、OCR

文本（例如评分）以及符号元素（例如图标或标志），任务是基于视觉上下文回答该查询。

MapQA（Li et al., 2025）：一个地理空间问答基准，使用 SQL 查询模板从 OpenStreetMap 构建。它包含覆盖九种推理类型（例如邻域推断、空间邻近性、类型分类）的问答对，并以真实世界地理实体几何为基础。

4.2 基线（Baseline）与指标（Metric）

MapAgent 是一个多智能体组合推理框架。因此，我们将其性能与智能体框架进行比较，例如 ReAct（Yao et al., 2023）和 LLMCompiler（Kim et al., 2024），以及与组合推理框架进行比较，例如 Chameleon（Lu et al., 2023）和 OctoTool（Lu et al., 2025）。对于 MapEval-API、MapEval-Textual 和 MapQA 数据集，我们使用 GPT-3.5-Turbo（？）和 Qwen-2.5-72b（？）。对于 MapEval-Visual 数据集，我们使用通过 vLLM 提供的模型，具体为 GPT-4o（？）和 Qwen-2.5-VL-72b（？）。我们使用准确率评估模型性能，以百分比表示。

4.3 结果

对 MapAgent 框架在四个不同且具有挑战性的地图相关数据集上的评估揭示了其在准确率方面的显著进步，持续优于现有的智能体方法，并展现出无论底层语言模型骨干如何都具有稳健的性能。这些发现强调了 MapAgent 作为广泛地图查询处理任务的稳健解决方案的有效性。

跨基准的显著提升 在所有评估的数据集上，MapAgent 框架始终实现高准确率。在表 2 的 MapEval-API 数据集上，MapAgent 在由 GPT-3.5-Turbo 和 Qwen-2.5-72B 驱动时，分别实现了约 70% 和 72% 的总体准确率。类似地，在表 3 的 MapEval-Textual 数据集上，MapAgent 使用 GPT-3.5-Turbo 达到 72.94% 的准确率，使用 Qwen-2.5-72B 达到 76.24%。表 5 中的 MapEval-Visual 结果进一步强化了这一趋势，MapAgent 使用 GPT-4o 实现了最高的总体准确率——68.95%，使用 Qwen-2.5-VL-72B 为 72.30%。最后，在表 6 的 MapQA 基准上，MapAgent 使用 GPT-3.5-Turbo 实现了 55.58% 的最高总体准确率，使用 Qwen-2.5-72B 为 55.93%。

超越先前方法 MapAgent 的性能在所有评估的数据集和语言模型上持续超越其他突出方法，包括 OctoTools、Chameleon、ReAct 和 LLMCompiler——尤其是表现第二好的 OctoTools。在图 5 中，在 MapEval-API 数据集上，MapAgent 在使用 GPT-3.5-Turbo 和 Qwen-2.5-72B 时，分别比 OctoTools 显示约 10% 和 11.54% 的准确率提升。对于 MapQA 基准，MapAgent 使用 GPT-3.5-Turbo 比 OctoTools 高出约 11.22%，使用

Qwen-2.5-72B 高出 9%。最后，在 MapEval-Textual 数据集上，MapAgent 的准确率
1301

LLM	MapQA										
	Method	Overall	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8	Type 9
GPT-3.5-Turbo	Base Model	24.35	32.04	7.43	5.98	50.56	7.22	31.44	18.34	10.14	56.04
	ReAct	20.88	34.02	7.80	5.44	58.02	6.02	30.13	12.14	9.88	24.44
	LLMCompiler	23.99	36.07	8.09	6.29	67.02	7.28	31.32	14.76	14.24	30.88
	Chameleon	38.11	47.52	17.59	20.77	73.14	15.52	40.12	14.28	43.68	70.40
	Octotools	44.74	50.00	20.40	22.00	90.67	18.52	43.78	41.20	48.24	67.83
	MapAgent	55.58	56.05	25.80	22.48	83.36	25.68	71.04	54.18	71.20	90.40
Qwen-2.5-72B	Base Model	23.51	29.60	8.40	7.20	48.00	7.60	30.00	17.20	9.20	54.40
	ReAct	21.69	30.80	9.20	7.60	51.60	9.40	32.80	16.40	8.60	28.80
	LLMCompiler	25.60	38.40	9.20	8.00	62.40	11.60	33.20	17.60	18.40	31.60
	Chameleon	40.00	51.20	27.60	23.20	68.80	22.40	38.40	20.40	38.80	68.40
	Octotools	46.18	54.39	30.91	25.80	85.67	24.44	41.60	38.20	43.24	71.33
	MapAgent	55.93	57.20	29.60	34.00	81.20	34.56	73.60	39.20	68.00	86.00

Table 6: Accuracy (%) of different methods and backbones on the MapQA Dataset.

Configuration	Google Map API	Our MapTools	Hierarchical Architecture	Accuracy (%)	Drop (%)
Raw API (Flat)	✓	✗	✗	39.37	49.00
Flat + MapTools	✓	✓	✗	56.39	27.18
MapAgent (Ours)	✓	✓	✓	77.44	-

Table 7: Ablation study of query-solving accuracy. We compare the full MapAgent against flattened architectures with and without the proposed MapTools .

is nearly 10% and 6% higher than OctoTools with GPT-3.5-Turbo and Qwen-2.5-72B. In the MapEval-Visual evaluations, MapAgent outperforms OctoTools by about 4.41% with GPT-4o and 2.8% with Qwen-2.5-VL-72B.

Backbone Agnostic A key strength of the MapAgent framework is its consistent high performance across different language model backbones. When evaluated on the MapEval-API and MapEval-Textual datasets, MapAgent demonstrates superior accuracy with both GPT-3.5-Turbo and Qwen-2.5-72B models. This is further evident in the MapEval-Visual dataset, where MapAgent achieves the highest accuracy with both GPT-4o and Qwen-2.5-VL-72B vision-language models. The consistent top-tier results across diverse LLMs and VLMs highlight MapAgent’s robustness and adaptability, showing its effectiveness is not tied to any specific model architecture.

5 Ablation Study: MapTools and Hierarchical Architecture

To evaluate the isolated impact of our proposed MapTools and the Hierarchical Architecture, we conducted an ablation study analyzing query solving accuracy across different configurations. To ensure a fair and computationally efficient comparison, this study was performed on a stratified subset

of the evaluation data, consisting of 50 randomly sampled data points from each of the four datasets (200 samples total). Consequently, the absolute accuracy reported here differs from the main results due to the smaller sample size.

We evaluated three settings: (1) the full system; (2) a flattened single-layer architecture using MapTools ; and (3) a flattened architecture where our specialized tools were replaced with direct Google Map API calls. The results are summarized in Table 7.

As shown in the table, when both our custom-developed tools and the hierarchical architecture are utilized, the solving accuracy on this subset reaches 77.44%. However, removing the hierarchical architecture causes a performance drop of 27.18%, lowering the accuracy to 56.39%. Furthermore, removing the MapTools (relying solely on the raw API in a flat architecture) results in an additional performance drop of 22%, with the final accuracy decreasing to 39.37%. These results demonstrate the critical significance of both the hierarchical reasoning structure and the specialized abstractions provided by MapTools .

For qualitative examples illustrating MapAgent behavior, refer to Appendix E.

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LLM										
MapQA										
方法	总体	类型 1	类型 2	类型 3	类型 4	类型 5	类型 6	类型 7	类型 8	类型 9
GPT-3.5-Turbo										
Base Model	24.35	32.04	7.43	5.98	50.56	7.22	31.44	18.34	10.14	56.04
ReAct	20.88	34.02	7.80	5.44	58.02	6.02	30.13	12.14	9.88	24.44
LLMCompiler	23.99	36.07	8.09	6.29	67.02	7.28	31.32	14.76	14.24	30.88
Chameleon	38.11	47.52	17.59	20.77	73.14	15.52	40.12	14.28	43.68	70.40
Octotools	44.74	50.00	20.40	22.00	90.67	18.52	43.78	41.20	48.24	67.83
MapAgent	55.58	56.05	25.80	22.48	83.36	25.68	71.04	54.18	71.20	90.40
Qwen-2.5-72B										
Base Model	23.51	29.60	8.40	7.20	48.00	7.60	30.00	17.20	9.20	54.40
ReAct	21.69	30.80	9.20	7.60	51.60	9.40	32.80	16.40	8.60	28.80
LLMCompiler	25.60	38.40	9.20	8.00	62.40	11.60	33.20	17.60	18.40	31.60
Chameleon	40.00	51.20	27.60	23.20	68.80	22.40	38.40	20.40	38.80	68.40
Octotools	46.18	54.39	30.91	25.80	85.67	24.44	41.60	38.20	43.24	71.33
MapAgent	55.93	57.20	29.60	34.00	81.20	34.56	73.60	39.20	68.00	86.00

表 6：不同方法和骨干模型在 MapQA 数据集上的准确率（%）。

配置 Google Map API我们的 MapTools分层架构准确率（%）下降（%）

原始 API（扁平）✓ x x 39.37 49.00

扁平 + MapTools ✓ ✓ x 56.39 27.18

MapAgent（我们的）✓ ✓ ✓ 77.44 -

表 7：查询求解准确率的消融研究。我们将完整的 MapAgent 与使用和不使用所提出的 MapTools 的扁平化架构进行比较。

在 GPT-3.5-Turbo 和 Qwen-2.5-72B 上，比 OctoTools 分别高出近 10% 和 6%。在 MapEval-Visual 评估中，MapAgent 在 GPT-4o 上比 OctoTools 高出约 4.41%，在 Qwen-2.5-VL-72B 上高出 2.8%。

骨干无关 MapAgent 框架的一个关键优势是其在不同的语言模型骨干上具有一致的高性能。在 MapEval-API 和 MapEval-Textual 数据集上评估时，MapAgent 在 GPT-3.5-Turbo 和 Qwen-2.5-72B 模型上都表现出更高的准确率。这在 MapEval-Visual 数据集中进一步得到体现，MapAgent 在 GPT-4o 和 Qwen-2.5-VL-72B 视觉语言模型上都取得了最高准确率。在不同 LLM 和 VLM 上一致的最高水平结果凸显了 MapAgent 的鲁棒性和适应性，表明其有效性并不依赖于任何特定的模型架构。

5 消融研究：MapTools 和分层架构

为了评估我们提出的 MapTools 和分层架构的单独影响，我们进行了一项消融研究，分析不同配置下的查询求解准确率。为了确保公平且计算高效的比较，本研究在评估数据的分层子集上进行，该子集由从四个数据集中各随机抽样的 50 个数据点组成（共 200 个样本）。因此，由于样本量较小，这里报告的绝对准确率与主要结果不同。

我们评估了三种设置：（1）完整系统；（2）使用 MapTools 的扁平化单层架构；（3）将我们的专用工具替换为直接 Google Map API 调用的扁平化架构。结果总结在表 7 中。

如表所示，当同时使用我们自定义开发的工具和分层架构时，该子集上的求解准确率达到 77.44%。然而，移除分层架构会导致性能下降 27.18%，准确率降至 56.39%。此外，移除 MapTools（在扁平架构中仅依赖原始 API）会导致额外的性能下降 22%，最终准确率降至 39.37%。这些结果证明了分层推理结构和 MapTools 提供的专用抽象的关键重要性。

关于说明 MapAgent 行为的定性示例，请参阅附录 E。

Dataset	Planner Agent Module Selection	Map-Tool Agent		Overall Error(%)
		Tools Calling	Parameter Passing	
MapEval-API	1.33	0.76	1.46	3.55
MapEval-Textual	0.67	1.02	0.93	2.62
MapQA	0.89	1.56	1.78	4.23
MapEval-Visual	2.5	0.66	1.2	4.36

Table 8: Detailed error analysis of MapAgent . We report the specific error rates (%) contributed by the Planner Agent (Module Selection) and the Map-Tool Agent (Tool Calling and Parameter Passing) across all four evaluation benchmarks.

6 Cost Analysis

We evaluate the computational efficiency of MapAgent against baselines Chameleon and Octotools across four key metrics: LLM API call frequency, total token consumption, Google Maps API usage, and end-to-end execution latency. Figure 3 summarizes the average resource consumption across all datasets, illustrating that MapAgent consistently achieves the lowest cost profile.

This efficiency stems from fundamental architectural differences between the systems. **Chameleon** relies on a sequential execution model, where solving a single query necessitates multiple distinct LLM API calls across various stages—including parameter extraction for individual modules (e.g., Nearby, Trip, Route), solution planning, and answer generation. This sequential dependency compounds both the API call count and overall latency. Similarly, **Octotools** employs an iterative, back-and-forth reasoning process. While flexible, this approach inherently generates additional LLM and Google Maps API interactions, resulting in higher average latency.

In contrast, MapAgent leverages a *parallel execution framework* within its inner Map-Tool Agent. By enabling the system to execute multiple tools simultaneously within a single LLM API call, MapAgent eliminates the need for separate interactions for each tool invocation. This architectural advantage dramatically reduces both the total number of API calls and the execution time, ensuring faster and more efficient query resolution across all evaluated configurations. Further statistical details regarding the cost analysis breakdown are provided in Appendix D.

7 Error Analysis and Challenges

To pinpoint the remaining challenges of MapAgent, we conducted a granular error analysis, examining specific error types at various stages of

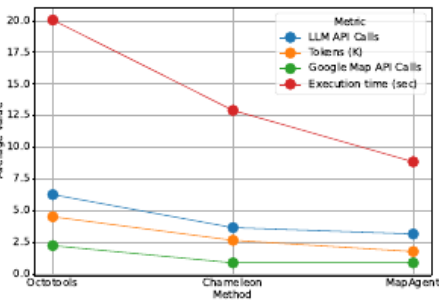


Figure 3: Aggregate cost analysis comparing MapAgent with Chameleon and Octotools. MapAgent achieves lower resource consumption across all four metrics: LLM API Calls, Token Usage, Google Maps API Calls, and Execution Time.

the framework. As illustrated in Figure 4, MapAgent exhibits an average error percentage across four geospatial benchmarks. While the average error rate is not significant, it stands at approximately 4.36% for MapEval-Visual datasets, 4.24% for MapQA, 3.55% for MapEval-API, and 2.62% for MapEval-Textual. Furthermore, Table 8 details the average percentage of errors attributed to each component within MapAgent. Our analysis reveals several key error sources:

Planner Agent: While generally accurate, the Planner Agent occasionally fails to select the appropriate modules from its inventory. We observed varying error rates depending on the task: approximately 0.67% for MapEval-Textual, 0.89% for MapQA, 1.33% for MapEval-API, and 2.5% for MapEval-Visual. These errors often arise from wrong component choices or operation order.

Map-Tool Agent: This agent can encounter two potential types of errors: fetching incorrect tools or passing improper arguments to those tools. While the error rate is not significant, it is detailed in Table 8.

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数据集 Planner Agent Map-Tool Agent 总体误差(%)模块选择 工具调用 参数传递

MapEval-API 1.33 0.76 1.46 3.55

MapEval-Textual 0.67 1.02 0.93 2.62

MapQA 0.89 1.56 1.78 4.23

MapEval-Visual 2.5 0.66 1.2 4.36

表 8 : MapAgent 的详细误差分析。我们报告了 Planner Agent (模块选择) 和 Map-Tool Agent (工具调用与参数传递) 在全部四个评估基准上贡献的具体错误率 (%) 。

6 成本分析

我们针对基线 Chameleon 和 Octotools , 在四个关键指标上评估了 MapAgent 的计算效率 : LLM API 调用频率、总 token 消耗量、Google Maps API 使用量以及端到端执行延迟。图 3 汇总了所有数据集上的平均资源消耗 , 表明 MapAgent 始终实现最低的成本概况。

这种效率源于系统之间根本性的架构差异。Chameleon

依赖顺序执行模型 , 解决单个查询需要在不同阶段进行多次不同的 LLM API 调用——包括为各个模块 (例如 Nearby、Trip、Route) 提取参数、解决方案规划和答案生成。这种顺序依赖关系同时加剧了 API 调用次数和整体延迟。类似地 , Octotools

采用迭代式、来回往复的推理过程。虽然灵活 , 但这种方法本质上会产生额外的 LLM 和 Google Maps API 交互 , 导致更高的平均延迟。

相比之下 , MapAgent 在其内部 Map-Tool Agent 中利用了并行执行框架。通过使系统能够在单次 LLM API 调用中同时执行多个工具 , MapAgent 消除了为每次工具调用进行单独交互的需要。这一架构优势大幅减少了 API 调用总数和执行时间 , 确保在所有评估配置下实现更快、更高效的查询解析。有关成本分析细分的进一步统计细节见附录 D。

7 误差分析与挑战

为了精准定位 MapAgent 剩余的挑战 , 我们进行了细粒度的误差分析 , 考察框架各个阶段的具体错误类型。

Octotools Chameleon MapAgent

方法

0.0

2.5

5.0

7.5

10.0

12.5

15.0

17.5

20.0平均值

指标

LLM API 调用

Token (K)

Google Map API 调用

执行时间 (秒)

图 3 : MapAgent 与 Chameleon 和 Octotools 的总体成本分析比较。MapAgent 在全部四个指标上实现了更低的资源消耗 : LLM API 调用、Token 使用量、Google Maps API 调用和执行时间。

如图 4 所示 , MapAgent 在四个地理空间基准上呈现出平均错误百分比。虽然平均错误率并不显著 , 但 MapEval-Visual 数据集约为 4.36% , MapQA 为 4.24% , MapEval-API 为 3.55% , MapEval-Textual 为

2.62%。此外，表 8 详细列出了归因于 MapAgent 内各组件的平均错误百分比。我们的分析揭示了几个关键错误来源：

Planner Agent：虽然总体准确，但 Planner Agent 偶尔无法从其清单中选择合适的模块。我们观察到不同任务的错误率有所差异：MapEval-Textual 约为 0.67%，MapQA 为 0.89%，MapEval-API 为 1.33%，MapEval-Visual 为 2.5%。这些错误通常源于错误的组件选择或操作顺序。

Map-Tool Agent：该 agent 可能遇到两种潜在类型的错误：获取错误的工具或向这些工具传递不恰当的参数。虽然错误率并不显著，但详见表 8。

1303

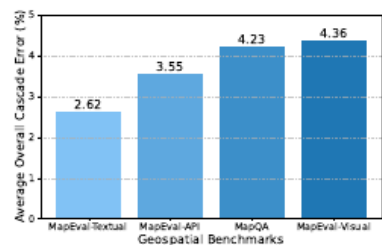


Figure 4: Average overall cascade error (%) for MapAgent across four geospatial benchmarks. The results demonstrate that the system maintains low error propagation rates ($< 4.5\%$) even on complex tasks like MapEval-Visual.

8 Broader Impact and Ethical Considerations

While MapAgent is designed for benign applications like urban navigation and location discovery, we recognize that advanced geospatial reasoning systems carry dual-use risks (Hovy et al., 2016).

Dual-Use Concerns. Our system’s ability to process ambiguous location descriptions and autonomously execute complex spatial queries could be misappropriated for surveillance or military purposes. The same capabilities that help users find restaurants or plan routes could theoretically be adapted to track individuals or support targeting systems. We strongly discourage deployments in contexts that violate privacy rights or contribute to harmful applications.

Data Bias and Equity. MapAgent relies on commercial map APIs that may contain inherent biases. If these sources underrepresent certain neighborhoods or communities, our system could inadvertently reinforce these gaps by directing users away from poorly documented areas. This creates risks of navigational inequity and unequal access to information. Developers should audit their data sources for bias and implement corrections where possible.

Privacy Considerations. Any deployment of geospatial reasoning systems should prioritize user privacy through informed consent, data minimization, and transparency about system capabilities. Users need to understand what information is collected and how their queries are processed. We recommend incorporating privacy-by-design principles and giving users control over their data.

We hope that by explicitly addressing these con-

cerns, we can encourage responsible development of geospatial AI systems while minimizing potential harms.

9 Conclusion

In this paper, we present MapAgent, a hierarchical multi-agent framework designed for effective geospatial reasoning. By decoupling planning from execution and introducing specialized modules with customized toolset—such as a dedicated map-tool agent—MapAgent addresses the limitations of flat agent-based architectures that struggle with tool overload and fine-grained API coordination. Through comprehensive evaluations on four challenging geospatial benchmarks—MapEval-Textual, MapEval-API, MapEval-Visual, and MapQA—MapAgent consistently outperforms existing tool-augmented and agentic baselines, including Chameleon and OctoTools. These results demonstrate MapAgent’s effectiveness and generalizability in real-world, map-integrated reasoning tasks. In future work, we plan to extend the framework to support multi-agent collaboration (Hosain et al., 2025) and to address broader multimodal spatio-temporal reasoning tasks.

Limitations

While our proposed hierarchical multi-agent framework, MapAgent, demonstrates substantial gains over state-of-the-art tool-augmented and agentic baselines for a variety of geospatial queries, it currently relies on a limited set of popular Google Maps APIs. Extending the framework to support a broader range of APIs and mapping services would further validate its generalizability and robustness. Additionally, although MapAgent features a plug-and-play architecture with the potential to be applied to other domains, such as web automation and software engineering, its effectiveness in these areas remains to be evaluated.

References

Sören Auer, Christian Bizer, Georgi Kobilarov, Jens Lehmann, Richard Cyganiak, and Zachary Ives. 2007. Dbpedia: A nucleus for a web of open data. In *international semantic web conference*, pages 722–735. Springer.

Zhenni Bi, Kai Han, Chuanjian Liu, Yehui Tang, and Yunhe Wang. 2024. Forest-of-thought: Scaling test-

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MapEval-Textual MapEval-API MapQA MapEval-Visual

Geospatial Benchmarks

0

1

2

3

4

5Average Overall Cascade Error (%)

2.62

3.55

4.23 4.36

Figure 4: MapAgent 在四个地理空间基准上的平均总体级联误差 (%)。结果表明，即使在 MapEval-Visual 等复杂任务上，该系统仍保持较低的误差传播率 (< 4.5%)。

8 更广泛的影响与伦理考量

尽管

MapAgent

是为城市导航和位置发现等良性应用而设计的，但我们认识到，先进的地理空间推理系统具有两用风险 (Hovy et al., 2016)。

两用关切。我们的系统处理模糊位置描述并自主执行复杂空间查询的能力，可能被滥用于监视或军事目的。那些帮助用户寻找餐厅或规划路线的相同能力，理论上也可以被改造用于追踪个人或支持瞄准系统。我们强烈反对在侵犯隐私权或助长有害应用的场景中部署该系统。

数据偏差与公平性。MapAgent 依赖商业地图 API，而这些 API 可能包含固有偏差。如果这些来源对某些街区或群体代表性不足，我们的系统可能会通过将用户引导远离记录不足的地区，无意中强化这些差距。这会造成导航不公平和信息获取不平等的风险。开发者应审查其数据源是否存在偏差，并在可能的情况下实施纠正。

隐私考量。任何地理空间推理系统的部署都应通过知情同意、数据最小化和系统能力透明化来优先保护用户隐私。用户需要了解收集了哪些信息以及他们的查询是如何被处理的。我们建议纳入隐私-by-design (privacy-by-design) 原则，并让用户控制自己的数据。

我们希望通过明确回应这些关切，能够鼓励地理空间 AI 系统的负责任开发，同时将潜在危害降至最低。

9 结论

在本文中，我们提出了 MapAgent，一个为有效地理空间推理而设计的分层多智能体 (multi-agent) 框架。通过将规划与执行解耦，并引入具有定制工具集的专门模块——例如专用的地图工具智能体——MapAgent

解决了扁平化基于智能体的架构所面临的局限，这些架构难以应对工具过载和细粒度

API

协调问题。通过对四个具有挑战性的地理空间基准——MapEval-Textual、MapEval-API、MapEval-Visual

和

MapQA——进行全面评估，MapAgent

始终优于现有的工具增强和智能体基线，包括

Chameleon

和

OctoTools。这些结果证明了 MapAgent 在真实世界、地图集成推理任务中的有效性和泛化能力。在未来工作中，我们计划扩展该框架以支持多智能体协作 (Hosain et al., 2025)，并处理更广泛的多模态时空推理任务。

局限性

尽管我们提出的分层多智能体框架

MapAgent

在各种地理空间查询上相较于最先进的工具增强和智能体基线表现出显著提升，但它目前依赖一组有限的主流

Google Maps API。将框架扩展到支持更广泛的 API 和地图服务，将进一步验证其泛化性和鲁棒性。此外，尽管 MapAgent

具有即插即用 (plug-and-play) 架构，并有可能应用于其他领域，例如

Web

自动化和软件工程，但其在这些领域中的有效性仍有待评估。

参考文献

Sören Auer, Christian Bizer, Georgi Kobilarov, Jens Lehmann, Richard Cyganiak, and Zachary Ives. 2007. Dbpedia:

A nucleus for a web of open data. In international semantic web conference, pages 722–735. Springer.

Zhenni Bi, Kai Han, Chuanjian Liu, Yehui Tang, and Yunhe Wang. 2024. Forest-of-thought: Scaling test-1304

- time compute for enhancing llm reasoning. *arXiv e-prints*, pages arXiv-2412.
- Kurt Bollacker, Robert Cook, and Patrick Tufts. 2007. Freebase: A shared database of structured general human knowledge. In *AAAI*, volume 7, pages 1962–1963.
- Nicholas J Car, Timo Homburg, Matthew Perry, John Herring, Frans Knibbe, Simon JD Cox, Joseph Abhayaratna, and Mathias Bonduel. 2022. Ogc geosparql: a geographic query language for rdf data. *OGC Implementation Standard*, *Open Geospatial Consortium*.
- Mahir Labib Dihan, Mohammed Eunus Ali, and Md Rizwan Parvez. 2025a. [MapQaTor: An extensible framework for efficient annotation of map-based QA datasets](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 1–10, Vienna, Austria. Association for Computational Linguistics.
- Mahir Labib Dihan, Md Tanvir Hassan, Md Tanvir Parvez, Md Hasebul Hasan, Md Almash Alam, Muhammad Aamir Cheema, Mohammed Eunus Ali, and Md Rizwan Parvez. 2025b. [Mapeval: A map-based evaluation of geo-spatial reasoning in foundation models](#). In *Forty-second International Conference on Machine Learning*.
- Yu Du, Fangyun Wei, and Hongyang Zhang. 2024. Anytool: Self-reflective, hierarchical agents for large-scale api calls. In *International Conference on Machine Learning*, pages 11812–11829. PMLR.
- Xinyu Guan, Li Lyna Zhang, Yifei Liu, Ning Shang, Youran Sun, Yi Zhu, Fan Yang, and Mao Yang. 2025. rstar-math: Small llms can master math reasoning with self-evolved deep thinking. *arXiv preprint arXiv:2501.04519*.
- Md Tanzib Hosain, Salman Rahman, Md Kishor Morol, and Md Rizwan Parvez. 2025. Xolver: Multi-agent reasoning with holistic experience learning just like an olympiad team. *arXiv preprint arXiv:2506.14234*.
- Dirk Hovy, Shannon L Spruit, and 1 others. 2016. The social impact of natural language processing. In *The 54th Annual Meeting of the Association for Computational Linguistics Proceedings of the Conference, Vol. 2 (Short Papers)*. Association for Computational Linguistics.
- Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2023. Swe-bench: Can language models resolve real-world github issues? *arXiv preprint arXiv:2310.06770*.
- Nikolaos Karalis, Georgios Mandilaras, and Manolis Koubarakis. 2019. Extending the yago2 knowledge graph with precise geospatial knowledge. In *The Semantic Web–ISWC 2019: 18th International Semantic Web Conference, Auckland, New Zealand, October 26–30, 2019, Proceedings, Part II* 18, pages 181–197. Springer.
- Sergios-Anestis Kefalidis, Dharmen Punjani, Eleni Tsalapati, Konstantinos Plas, Mariangela Pollali, Michail Mitsios, Myrto Tsokanaridou, Manolis Koubarakis, and Pierre Maret. 2023. Benchmarking geospatial question answering engines using the dataset geoquestions1089. In *International semantic web conference*, pages 266–284. Springer.
- Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish Sabharwal. 2023. [Decomposed prompting: A modular approach for solving complex tasks](#). In *The Eleventh International Conference on Learning Representations*.
- Sehoon Kim, Suhong Moon, Ryan Tabrizi, Nicholas Lee, Michael W Mahoney, Kurt Keutzer, and Amir Gholami. 2024. An llm compiler for parallel function calling. In *Forty-first International Conference on Machine Learning*.
- Zekun Li, Malcolm Grossman, Mihir Kulkarni, Muhao Chen, Yao-Yi Chiang, and 1 others. 2025. Mapqa: Open-domain geospatial question answering on map data. *arXiv preprint arXiv:2503.07871*.
- Pan Lu, Bowen Chen, Sheng Liu, Rahul Thapa, Joseph Boen, and James Zou. 2025. [Octotools: An agentic framework with extensible tools for complex reasoning](#). In *ICLR 2025 Workshop on Foundation Models in the Wild*.
- Pan Lu, Baolin Peng, Hao Cheng, Michel Galley, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, and Jianfeng Gao. 2023. Chameleon: Plug-and-play compositional reasoning with large language models. *Advances in Neural Information Processing Systems*, 36:43447–43478.
- Gengchen Mai, Weiming Huang, Jin Sun, Suhang Song, Deepak Mishra, Ninghao Liu, Song Gao, Tianming Liu, Gao Cong, Yingjie Hu, Chris Cundy, Ziyuan Li, Rui Zhu, and Ni Lao. 2024. [On the opportunities and challenges of foundation models for geoai \(vision paper\)](#). *ACM Trans. Spatial Algorithms Syst.*, 10(2).
- Md Rizwan Parvez. 2025. [Chain of evidences and evidence to generate: Prompting for context grounded and retrieval augmented reasoning](#). In *Proceedings of the 4th International Workshop on Knowledge-Augmented Methods for Natural Language Processing*, pages 230–245, Albuquerque, New Mexico, USA. Association for Computational Linguistics.
- Ethan Perez, Patrick Lewis, Wen-tau Yih, Kyunghyun Cho, and Douwe Kiela. 2020. Unsupervised question decomposition for question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8864–8880.

中文翻译 - 对应原文第 10 页

用于增强 LLM 推理的测试时计算。arXiv

e-prints , 页码 arXiv-2412。

Kurt Bollacker, Robert Cook, and Patrick Tufts. 2007.

Freebase : 一个结构化通用人类知识的共享数据库。In AAAI, volume 7, pages 1962–1963。

Nicholas J Car, Timo Homburg, Matthew Perry, John

Herring, Frans Knibbe, Simon JD Cox, Joseph Abha-

yaratna, and Mathias Bonduel. 2022. Ogc geosparql-

一种用于 rdf 数据的地理查询语言。OGC Im-

plementation Standard., Open Geospatial Consor-

tium。

Mahir Labib Dihan, Mohammed Eunus Ali, and

Md Rizwan Parvez. 2025a. MapQaTor : 一个用于高效标注基于地图的

QA 数据集的可扩展框架。In Proceedings of the 63rd Annual

Meeting of the Association for Computational Lin-

guistics (Volume 3: System Demonstrations), pages 1–

10, Vienna, Austria. Association for Computational

Linguistics。

Mahir Labib Dihan, Md Tanvir Hassan, Md Tanvir

Parvez, Md Hasebul Hasan, Md Almash Alam,

Muhammad Aamir Cheema, Mohammed Eunus Ali,

and Md Rizwan Parvez. 2025b. Mapeval : 基础模型中基于地图的

地理空间推理评估。In Forty-second International Confer-

ence on Machine Learning。

Yu Du, Fangyun Wei, and Hongyang Zhang. 2024. Any-

tool : 用于大规模 api 调用的自反思、分层智能体。In International Conference on Ma-

chine Learning, pages 11812–11829. PMLR。

Xinyu Guan, Li Lyna Zhang, Yifei Liu, Ning Shang,

Youran Sun, Yi Zhu, Fan Yang, and Mao Yang.

2025. rstar-math : 小型 llm 可以通过自演化深度思考掌握数学推

理。arXiv preprint

arXiv:2501.04519。

Md Tanzib Hosain, Salman Rahman, Md Kishor Morol,

and Md Rizwan Parvez. 2025. Xolver : 像

奥林匹克团队一样进行整体经验学习的多智能体

推理。arXiv preprint arXiv:2506.14234。

Dirk Hovy, Shannon L Spruit, and 1 others. 2016. 自然语言处理的

社会影响。In The

54th Annual Meeting of the Association for Compu-

tational Linguistics Proceedings of the Conference,

Vol. 2 (Short Papers). Association for Computational

Linguistics。

Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2023. Swe-bench : 语言模型能解决真实世界的 github 问题吗? arXiv preprint arXiv:2310.06770。

Nikolaos Karalis, Georgios Mandilaras, and Manolis Koubarakis. 2019. 用精确地理空间知识扩展 yago2 知识图谱。In The Semantic Web—ISWC 2019: 18th International Semantic Web Conference, Auckland, New Zealand, October 26–30, 2019, Proceedings, Part II 18 , pages 181–197. Springer。

Sergios-Anestis Kefalidis, Dharmen Punjani, Eleni Tsalapati, Konstantinos Plas, Mariangela Pollali, Michail Mitsios, Myrto Tsokanaridou, Manolis Koubarakis, and Pierre Maret. 2023. 使用数据集 geoquestions1089 对地理空间问答引擎进行基准测试。In International semantic web conference, pages 266–284. Springer。

Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish Sabharwal. 2023. 分解式提示 : 解决复杂任务的模块化方法。In The Eleventh International Conference on Learning Representations。

Sehoon Kim, Suhong Moon, Ryan Tabrizi, Nicholas Lee, Michael W Mahoney, Kurt Keutzer, and Amir Gholami. 2024. 用于并行函数调用的 llm 编译器。In Forty-first International Conference on Machine Learning。

Zekun Li, Malcolm Grossman, Mihir Kulkarni, Muhao Chen, Yao-Yi Chiang, and 1 others. 2025. Mapqa : 基于地图数据的开放域地理空间问答。arXiv preprint arXiv:2503.07871。

Pan Lu, Bowen Chen, Sheng Liu, Rahul Thapa, Joseph Boen, and James Zou. 2025. Octotools : 一个用于复杂推理的、具有可扩展工具的智能体框架。In ICLR 2025 Workshop on Foundation Models in the Wild。

Pan Lu, Baolin Peng, Hao Cheng, Michel Galley, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, and Jianfeng Gao. 2023. Chameleon : 使用大型语言模型的即插即用组合式推理。Advances in Neural Information Processing Systems , 36:43447–43478。

Gengchen Mai, Weiming Huang, Jin Sun, Suhan Song, Deepak Mishra, Ninghao Liu, Song Gao, Tianming

Liu, Gao Cong, Yingjie Hu, Chris Cundy, Ziyuan Li, Rui Zhu, and Ni Lao. 2024. 论基础模型用于 geoai 的机遇与挑战 (愿景论文) 。ACM Trans. Spatial Algorithms Syst., 10(2)。

Md Rizwan Parvez. 2025. 证据链与证

据生成：面向上下文接地
和检索增强推理的提示。In Proceedings
of the 4th International Workshop on Knowledge-
Augmented Methods for Natural Language Process-
ing, pages 230–245, Albuquerque, New Mexico,
USA. Association for Computational Linguistics。

Ethan Perez, Patrick Lewis, Wen-tau Yih, Kyunghyun
Cho, and Douwe Kiela. 2020. 用于问答的无监督问题
分解。In Proceed-
ings of the 2020 Conference on Empirical Methods
in Natural Language Processing (EMNLP) , pages
8864–8880。
1305

- Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, and 1 others. 2023. Toolllm: Facilitating large language models to master 16000+ real-world apis. *arXiv preprint arXiv:2307.16789*.
- Jonathan Roberts, Timo Lüddecke, Sowmen Das, Kai Han, and Samuel Albanie. 2023. [GPT4GEO: How a language model sees the world's geography](#). In *NeurIPS 2023 Foundation Models for Decision Making Workshop*.
- Fabian M Suchanek, Gjergji Kasneci, and Gerhard Weikum. 2007. Yago: a core of semantic knowledge. In *Proceedings of the 16th international conference on World Wide Web*, pages 697–706.
- Simeng Sun, Yang Liu, Shuohang Wang, Dan Iter, Chenguang Zhu, and Mohit Iyyer. 2024. Pearl: Prompting large language models to plan and execute actions over long documents. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 469–486.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.
- Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V Le, and Ed H. Chi. 2023. [Least-to-most prompting enables complex reasoning in large language models](#). In *The Eleventh International Conference on Learning Representations*.

中文翻译 - 对应原文第 11 页

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, and 1 others. 2023. Toolllm: Facilitating large language models to master 16000+ real-world apis. arXiv preprint arXiv:2307.16789.

Jonathan Roberts, Timo Lüddecke, Sowmen Das, Kai Han, and Samuel Albanie. 2023. GPT4GEO: How a language model sees the world's geography. In NeurIPS 2023 Foundation Models for Decision Making Workshop.

Fabian M Suchanek, Gjergji Kasneci, and Gerhard Weikum. 2007. Yago: a core of semantic knowledge. In Proceedings of the 16th international conference on World Wide Web, pages 697–706.

Simeng Sun, Yang Liu, Shuohang Wang, Dan Iter, Chenguang Zhu, and Mohit Iyyer. 2024. Pearl: Prompting large language models to plan and execute actions over long documents. In Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), pages 469–486.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. Chain-of-thought prompting elicits reasoning in large language models. Advances in neural information processing systems, 35:24824–24837.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. In International Conference on Learning Representations (ICLR).

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V Le, and Ed H. Chi. 2023. Least-to-most prompting enables complex reasoning in large language models. In The Eleventh International Conference on Learning Representations.

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A How MapAgent Outperforms Chameleon and OctoTools

Appendices A.1, A.2, and A.3 illustrate an example where OctoTools and Chameleon failed to solve a complex geospatial query, but MapAgent could. The main reason behind this failure is that both OctoTools and Chameleon have all their module or tool cards in one layer. Consequently, they have trouble guessing the exact tool/module for fetching information. The example illustrates that solving this query requires the distance and travel time from one location to another for each location in the query, which a trip tool can deliver correctly. However, both Chameleon and OctoTools struggle here and select a route tool to fetch the information, subsequently lacking information about all the routes between all the places. Additionally, OctoTools encounters another limitation. As shown in Table 9, where the intermediate reasoning steps of OctoTools are displayed, we found that this framework hallucinates and triggers a tool with parameters that have no relation to this question, which yields wrong information. These limitations trigger a failure to provide all the required information to the subsequent steps. However, MapAgent can effectively solve this by introducing an inner layer in the Google Map module. This module, communicating with a map-tool agent, appropriately triggers the trips tool and retrieves the relevant information that is sufficient for the subsequent steps to determine the solutions.

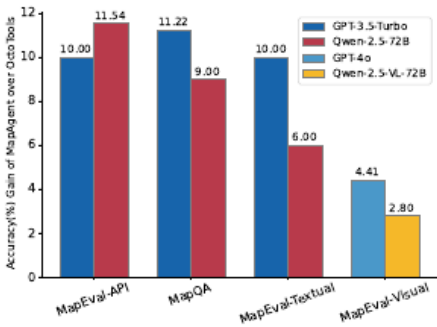


Figure 5: Accuracy improvements of MapAgent over OctoTools on various benchmarks.

A.1 Octotools Output **Green: Correct Answer** **Red: Wrong Answer**

- PID: 19

- **Query:** I am at Rydges Wellington. Starting from my current location, I will first go to Kelburn Park, followed by a trip to Victoria University of Wellington, and finally return to my current location. All the travel will be done using my own car. How much time will the entire journey take?
- **Options:** A.11 mins B.10 mins C.12 mins D.9 mins
- **Correct Answer:** B
- **Reasoning Steps:** Table 9 shows the reasoning steps of OctoTools and the tools it executes in each step. Within the table, **red** highlights indicate hallucinations, and **orange** highlights denote incorrect information that OctoTools attempts to fetch after execution. We observe that the initial call to the Route Tool, using the same origin and destination, leads to inaccurate information retrieval. Furthermore, the subsequent nine calls, shown in table rows 2 through 5, demonstrate hallucination and the use of parameters unrelated to the question, revealing a lack of correct tool calling. Although a trip tool capable of providing all relevant information in a single call exists, it is not executed correctly.

- **Octotools Final Output:** To determine the total travel time for the round trip starting and ending at Rydges Wellington, with stops at Kelburn Park and Victoria University of Wellington, the following steps were taken:

Step 1: Analyze the Query The query specifies a round trip: 1. Start at **Rydges Wellington**. 2. Visit **Kelburn Park**. 3. Proceed to **Victoria University of Wellington**. 4. Return to **Rydges Wellington**. The mode of transport is driving.

Step 2: Use the Route Tool The **Route Tool** was used to calculate the total driving time for the specified route. The tool provided the following result: - **Total Route Distance**: 1.9 km. - **Total Driving Time**: 9 minutes.

Step 3: Verify the Route Details The route details confirm the driving time includes all segments of the trip: - Rydges Wellington → Kelburn Park. - Kelburn Park → Victoria University of Wellington. - Victoria University of Wellington → Rydges Wellington.

A MapAgent 如何优于 Chameleon 和 OctoTools

附录 A.1、A.2 和 A.3 展示了一个示例，其中 OctoTools 和 Chameleon 未能解决一个复杂的地理空间查询，但 MapAgent 可以。这种失败背后的主要原因是 OctoTools 和 Chameleon 将所有模块或工具卡片都放在一层中。因此，它们难以猜测用于获取信息的准确工具/模块。该示例表明，解决此查询需要查询中每个位置到另一位置的距离和旅行时间，而行程工具 (trip tool) 可以正确提供这些信息。然而，Chameleon 和 OctoTools 在这里都遇到了困难，并选择路线工具 (route tool) 来获取信息，随后缺乏所有地点之间所有路线的信息。此外，OctoTools 还遇到了另一个限制。如表 9 所示，其中显示了 OctoTools 的中间推理步骤，我们发现该框架产生了幻觉，并触发了一个带有与问题无关参数的工具，从而产生了错误信息。这些限制导致无法向后续步骤提供所有所需信息。然而，MapAgent 可以通过在 Google Map 模块中引入一个内层来有效解决这个问题。该模块与一个地图工具智能体 (map-tool agent) 通信，适当地触发行程工具并检索相关信息，这些信息足以让后续步骤确定解决方案。

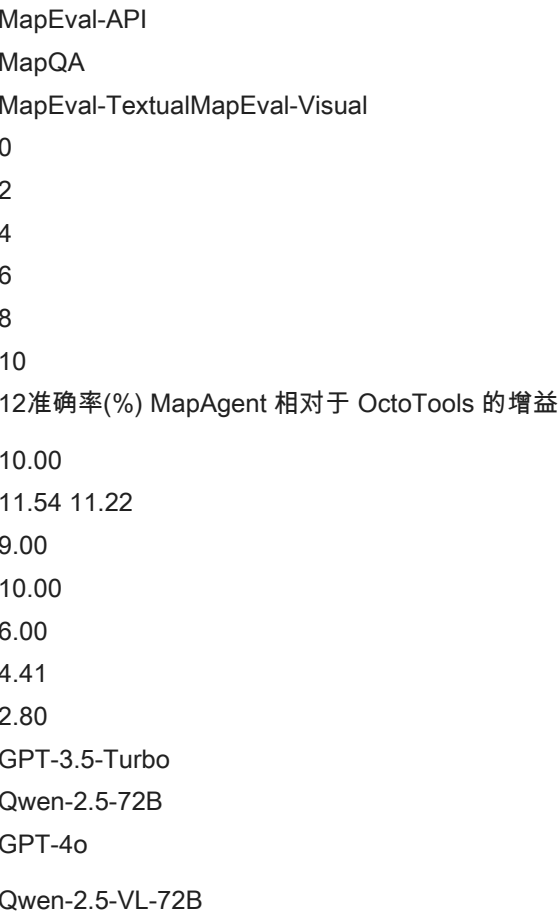


图 5：MapAgent 相对于 OctoTools 在各种基准测试上的准确率提升。

A.1 Octotools 输出 绿色：正确答案 红色：错误答案

- PID : 19
- 查询：我在 Rydges Wellington。从我的当前位置出发，我将首先前往 Kelburn Park，然后前往 Victoria University of Wellington，最后返回我的当前位置。所有行程都将使用我自己的汽车完成。整个旅程需要多长时间？
- 选项：A.11 分钟 B.10 分钟 C.12 分钟 D.9 分钟
- 正确答案：B
- 推理步骤：表 9 显示了 OctoTools

的推理步骤及其在每个步骤中执行的工具。在表中，红色高亮表示幻觉，橙色高亮表示 OctoTools 在执行后试图获取的不正确信息。我们观察到，首次调用 Route Tool 时，使用相同的起点和终点，导致信息检索不准确。此外，表第 2 行至第 5 行中显示的后续九次调用表明存在幻觉，并使用了与问题无关的参数，揭示出缺乏正确的工具调用。尽管存在一个能够在单次调用中提供所有相关信息的行程工具，但它没有被正确执行。

- Octotools 最终输出：为了确定从 Rydges Wellington 出发并返回 Rydges Wellington、途中停靠 Kelburn Park 和 Victoria University of Wellington 的往返总旅行时间，采取了以下步骤：
步骤 1：分析查询 查询指定了一次往返行程：1. 从 Rydges Wellington 出发。2. 访问 Kelburn Park。3. 前往 Victoria University of Wellington。4. 返回 Rydges Wellington。交通方式为驾车。
步骤 2：使用 Route Tool 使用 Route Tool 计算指定路线的总驾车时间。该工具提供了以下结果：- 总路线距离：1.9 公里。- 总驾车时间：9 分钟。
步骤 3：验证路线详情 路线详情确认驾车时间包括行程的所有路段：- Rydges Wellington → Kelburn Park。- Kelburn Park → Victoria University of Wellington。- Victoria University of Wellington → Rydges Wellington。

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Conclusion: The total driving time for the entire journey is **9 minutes**.

Final Answer: D. 9 mins

A.2 Chameleon Output **Green: Correct**

Answer **Red** Wrong Answer

- **PID:** 19
- **Query:** I am at Rydges Wellington. Starting from my current location, I will first go to Kelburn Park, followed by a trip to Victoria University of Wellington, and finally return to my current location. All the travel will be done using my own car. How much time will the entire journey take?
- **Options:** A.11 mins B.10 mins C.12 mins D.9 mins
- **Correct Answer:** B
- **Selected Module by the Planner:** Route, solution_generator, answer_generator
- **Information gathered by Route Module:** There are total 1 routes from Rydges Wellington, Wellington 6011, New Zealand to Victoria University of Wellington, Wellington 6012, New Zealand. The route information is provided below:
Route 1:(VIA) Parkvale Rd (7.6 km, 20 mins). Details steps are provided below:
Head southwest on Featherston St toward Whitmore St (35 m, 1 min)
Turn right at the 1st cross street onto Whitmore St (0.2 km, 1 min)
Continue onto Bowen St (0.8 km, 2 mins)
Turn left onto Tinakori Rd (signs for Karori/Botanic Garden) (0.2 km, 1 min)
Continue onto Glenmore St (1.6 km, 3 mins)
At the roundabout, take the 1st exit and stay on Glenmore St (0.2 km, 1 min)
Continue onto Karori Tunnel (0.1 km, 1 min)
Continue onto Chaytor St (0.9 km, 2 mins)
Continue onto Karori Rd (1.2 km, 2 mins)
Turn right onto Parkvale Rd (1.1 km, 2 mins)
Turn right to stay on Parkvale Rd (0.3 km, 1 min)
Turn right to stay on Parkvale Rd (1.1 km, 5 mins)
- **Solution Generated by Solution Generator:** It seems there's a mismatch in the information

provided. The detailed route you've given is for travel between Rydges Wellington and Victoria University of Wellington, taking approximately 20 minutes. You haven't provided a detailed route or travel time for the leg between Rydges Wellington and Kelburn Park, or for the return trip from Victoria University of Wellington back to Rydges Wellington via Kelburn Park. Based solely on the 20-minute travel time you just provided for Rydges to Victoria University (Route 1), we cannot accurately answer the question about the entire journey involving Kelburn Park without knowing the travel times for the other segments. However, if we were to ignore the previous travel times you mentioned and assume the question implies a direct trip from Rydges to Victoria University and back, using the 20-minute route in each direction, the total would be 20 minutes + 20 minutes = 40 minutes. This doesn't align with the provided options. So, the answer should be E.

• Answer Generator Module Output:

Not able to answer the question

A.3 MapAgent Output **Green: Correct**

Answer **Red** Wrong Answer

- **PID:** 19
- **Query:** I am at Rydges Wellington. Starting from my current location, I will first go to Kelburn Park, followed by a trip to Victoria University of Wellington, and finally return to my current location. All the travel will be done using my own car. How much time will the entire journey take?
- **Options:** A.11 mins B.10 mins C.12 mins D.9 mins
- **Correct Answer:** B
- **Selected Module by the Planner Agent:** map_service, solution_generator, answer_generator
- **Information gathered by Google_Maps Module:**
All Location Info:
Rydges Wellington
Name: Rydges Wellington
Address: 75 Featherston Street, Pipitea, Wellington 6011, New Zealand

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结论：整个行程的总驾驶时间为 9 分钟。

最终答案：D. 9 mins

A.2 Chameleon 输出 绿色：正确答案 红色：错误答案

• PID：19

• 查询：我在 Rydges Wellington。从我的当前位置出发，我将首先前往 Kelburn Park，随后前往 Victoria University of Wellington，最后返回我的当前位置。所有出行都将使用我自己的汽车完成。整个行程需要多长时间？

• 选项：A.11 mins B.10 mins C.12 mins D.9 mins

• 正确答案：B

• Planner 选择的模块：Route、solution_generator、answer_generator

• Route 模块收集的信息：

从 Rydges Wellington, Wellington 6011, New Zealand 到 Victoria University of Wellington, Wellington 6012, New Zealand 共有 1 条路线。路线信息如下：

路线 1：（经由）Parkvale Rd（7.6 km，20 mins）。

详细步骤如下：

沿 Featherston St 向西南方向行驶，朝 Whitmore St 前进（35 m，1 min）

在第 1 个交叉路口右转，进入 Whitmore St（0.2 km，1 min）

继续进入 Bowen St（0.8 km，2 mins）

左转进入 Tinakori Rd（指示牌指向 Karori/Botanic Garden）（0.2 km，1 min）

继续进入 Glenmore St（1.6 km，3 mins）

在环岛处，从第 1 个出口驶出并继续沿 Glenmore St 行驶（0.2 km，1 min）

继续进入 Karori Tunnel（0.1 km，1 min）

继续进入 Chaytor St（0.9 km，2 mins）

继续进入 Karori Rd（1.2 km，2 mins）

右转进入 Parkvale Rd（1.1 km，2 mins）

右转以继续沿 Parkvale Rd 行驶（0.3 km，1 min）

右转以继续沿 Parkvale Rd 行驶（1.1 km，5 mins）

• Solution Generator 生成的解决方案：

所提供的信息似乎存在不匹配。你给出的详细路线是 Rydges Wellington 与 Victoria University of Wellington 之间的行程，耗时约 20 分钟。你没有提供 Rydges Wellington 与 Kelburn Park 之间那一段的详细路线或旅行时间，也没有提供从 Victoria University of Wellington 经 Kelburn Park 返回 Rydges Wellington 的返程信息。仅根据你刚刚提供的 Rydges 到 Victoria University 的 20 分钟旅行时间（路线 1），在不知道其他路段旅行时间的情况下，我们无法准确回答涉及 Kelburn Park 的整个行程问题。

然而，如果我们忽略你提到的先前旅行时间，并假设问题暗示的是从 Rydges 到 Victoria University 再返回的直接行程，且每个方向都使用 20 分钟路线，那么总计将是 20 分钟 + 20 分钟 = 40 分钟。

这与提供的选项不一致。

所以，答案应为 E。

• Answer Generator 模块输出：

无法回答该问题

A.3 MapAgent 输出 绿色：正确答案 红色：错误答案

• PID：19

• 查询：我在 Rydges Wellington。从我的当前位置出发，我将首先前往 Kelburn Park，随后前往 Victoria University of Wellington，最后返回我的当前位置。所有出行都将使用我自己的汽车完成。整个行程需要多长时间？

• 选项：A.11 mins B.10 mins C.12 mins D.9 mins

- 正确答案：B
- Planner Agent 选择的模块：
map_service、solution_generator、answer_generator
- Google_Maps 模块收集的信息：
所有位置信息：

Rydges Wellington

名称：Rydges Wellington

地址：75 Featherston Street, Pipitea, Wellington 6011, New Zealand
1308

Rating: 4.2
 Types: lodging, point_of_interest, establishment
 Is Open Now: True
 Weekday Opening Hours:
 - Monday: Open 24 hours
 - Tuesday: Open 24 hours
 - Wednesday: Open 24 hours
 - Thursday: Open 24 hours
 - Friday: Open 24 hours
 - Saturday: Open 24 hours
 - Sunday: Open 24 hours
 Kelburn Park
 Name: Kelburn Park
 Address: Salamanca Road, Kelburn, Wellington 6012, New Zealand
 Rating: 4.4
 Types: park, tourist_attraction, point_of_interest, establishment
 Is Open Now: True
 Weekday Opening Hours:
 - Monday: Open 24 hours
 - Tuesday: Open 24 hours
 - Wednesday: Open 24 hours
 - Thursday: Open 24 hours
 - Friday: Open 24 hours
 - Saturday: Open 24 hours
 - Sunday: Open 24 hours
 Victoria University of Wellington
 Name: Victoria University of Wellington
 Address: Kelburn Parade, Kelburn, Wellington 6012, New Zealand
 Rating: 4.3
 Types: university, point_of_interest, establishment
 Is Open Now: N/A
 Weekday Opening Hours:
 - Unknown
 The travel time(distance) from Rydges Wellington to Kelburn Park is 5 mins (1.6 km)
 The travel time(distance) from Rydges Wellington to Victoria University of Wellington is 1 min (1 m)
 The travel time(distance) from Kelburn Park to Rydges Wellington is 4 mins (1.4 km)
 The travel time(distance) from Kelburn Park to Victoria University of Wellington is 4 mins (1.4 km)
 The travel time(distance) from Victoria University of Wellington to Rydges Wellington is 1 min (1 m)
 The travel time(distance) from Victoria

University of Wellington to Kelburn Park is 5 mins (1.6 km)

• Solution Generated by Solution Generator

Module: To calculate the total time for the entire journey, we need to consider the travel time from Rydges Wellington to Kelburn Park, then from Kelburn Park to Victoria University of Wellington, and finally the return trip from Victoria University of Wellington to Rydges Wellington.

The travel time from Rydges Wellington to Kelburn Park is 5 minutes. The travel time from Kelburn Park to Victoria University of Wellington is 4 minutes. The return trip from Victoria University of Wellington to Rydges Wellington is 1 minutes.

Adding these times together, the total time for the entire journey is $5 + 4 + 1 = 10$ minutes. So, the answer is 10 minutes. The most possible option is "the answer is B"

• Answer Generated by Answer Generator

Module: B. 10 mins

B Configurations in MapAgent

B.1 Map-Tool Agent

Prompt for Map-Tool Agent

You are an agent designed to understand user questions and retrieve relevant information using specific tools. When a user asks a question, your task is to identify the appropriate tool from the available list (Trip Tool, Route Tool, Nearby, and PlaceInfo) that can best answer the query. You will then use that tool to fetch the information and provide the retrieved data to the user. You are not expected to answer the question directly; your role is solely to retrieve the necessary information using the designated tools. For instance, if you think you need to use more than one tool, you can use them in parallel.

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评分：4.2
类型：住宿、兴趣点、机构
当前是否营业：是
工作日营业时间：

- 星期一：24 小时营业
- 星期二：24 小时营业
- 星期三：24 小时营业
- 星期四：24 小时营业
- 星期五：24 小时营业
- 星期六：24 小时营业
- 星期日：24 小时营业

Kelburn Park
名称：Kelburn Park
地址：Salamanca Road, Kelburn, Wellington 6012, New Zealand

评分：4.4
类型：公园、旅游景点、兴趣点、机构
当前是否营业：是
工作日营业时间：

- 星期一：24 小时营业
- 星期二：24 小时营业
- 星期三：24 小时营业
- 星期四：24 小时营业
- 星期五：24 小时营业
- 星期六：24 小时营业
- 星期日：24 小时营业

Victoria University of Wellington
名称：Victoria University of Wellington
地址：Kelburn Parade, Kelburn, Wellington 6012, New Zealand

评分：4.3
类型：大学、兴趣点、机构
当前是否营业：N/A
工作日营业时间：

- 未知

从 Rydges Wellington 到 Kelburn Park 的旅行时间（距离）为 5 分钟（1.6 km）
从 Rydges Wellington 到 Victoria University of Wellington 的旅行时间（距离）为 1 分钟（1 m）
从 Kelburn Park 到 Rydges Wellington 的旅行时间（距离）为 4 分钟（1.4 km）
从 Kelburn Park 到 Victoria University of Wellington 的旅行时间（距离）为 4 分钟（1.4 km）
从 Victoria University of Wellington 到 Rydges Wellington 的旅行时间（距离）为 1 分钟（1 m）
从 Victoria University of Wellington 到 Kelburn Park 的旅行时间（距离）为 5 分钟（1.6 km）

• 由 Solution Generator 模块生成的解决方案：要计算整个行程的总时间，我们需要考虑从 Rydges Wellington 到 Kelburn Park 的旅行时间，然后是从 Kelburn Park 到 Victoria University of Wellington 的旅行时间，最后是从 Victoria University of Wellington 返回 Rydges Wellington 的返程时间。

从 Rydges Wellington 到 Kelburn Park 的旅行时间是 5 分钟。从 Kelburn Park 到 Victoria University of Wellington 的旅行时间是 4 分钟。从 Victoria University of Wellington 返回 Rydges Wellington 的返程时间是 1 分钟。
将这些时间相加，整个行程的总时间是 $5 + 4 + 1 = 10$ 分钟。

所以，答案是 10 分钟。最可能的选项是“答案是 B”

- 由 Answer Generator 模块生成的答案：B. 10 mins

B MapAgent 中的配置

B.1 Map-Tool Agent

Map-Tool Agent 的提示词

你是一个

agent，旨在理解用户问题并使用特定工具检索相关信息。当用户提出问题时，你的任务是从可用列表 (Trip

Tool、Route Tool、Nearby 和

PlaceInfo) 中识别出最能回答该查询的合适工具。然后你将使用该工具获取信息，并将检索到的数据提供给用户。

你不应该直接回答问题；你的角色仅仅是使用指定工具检索必要信息。例如，如果你认为需要使用多个工具，可以并行使用它们。

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B.2 Solution Generator (MapEval-API)

Prompt for Solution Generator

Given the question (and the context), select the answer from the options ["A", "B", "C", "D"]. You should give concise and step-by-step solutions. Finally, conclude the answer in the format of "the answer is [ANSWER]", where [ANSWER] is one from the options ["A", "B", "C", "D"]. For example, "the answer is A", "the answer is B", "the answer is C", or "the answer is D". If the answer is not in the options, declare it as Unanswerable.

Example 1

Question: What is the address of Multiplan Center?

Options: (A) 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh (B) 38/1/C BC DAS Street Lalbagh (C) Polashi,BUET (D) Central Road, USA

Metadata: {"skill": "Fetch context from corresponding google map api and based on the context answer the question"}

Google Maps response:
Name: Multiplan Center
Address: 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh
Rating: 4.4
Types: point_of_interest, establishment
Is Open Now: False
Weekday Opening Hours:
- Monday: 10:00AM–8:00PM
- Tuesday: Closed
- Wednesday: 10:00AM–8:00PM
- Thursday: 10:00AM–8:00PM
- Friday: 10:00AM–8:00PM
- Saturday: 10:00AM–8:00PM
- Sunday: 10:00AM–8:00PM

Solution: If you look at the context and search, then after reaching The address of Multiplan Center is 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh. Therefore, the answer is B.

.... <4 more examples>

Now Answer the question following.

Question: {question}

B.3 Visual Place Recognizer

Prompt for Visual Place Recognizer

I will provide you with an image. You must determine the precise center location within the image and a prediction of the boundary of the image by predicting a radius from the center. Your return format should be the center location name followed by a space, then the radius. Based on that, provide only one complete address in a single, consistent format followed by a space and a radius. Do not include any extraneous text before or after the address.

C Detailed Tool Parameters

The following section outlines the tools utilized by the Map Service module, as retrieved by the map-tool agent. Each tool is defined along with its input parameters.

Trip Tool

current_location (str): The starting location of the trip.

visiting_places (list): A list of locations to visit.

travel_mode (str): The mode of travel, defaults to driving.

Route Tool

origin (str): The starting location for the directions.

destination (str): The destination for the directions.

travel_mode (str): The mode of travel, defaults to driving.

mode (str): The mode of transportation to use for the directions, such as "driving"

alternatives (bool): Whether to return multiple possible routes.

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B.2 解决方案生成器 (MapEval-API)

解决方案生成器的提示词

给定问题 (以及上下文) , 从选项 ["A"、"B"、"C"、"D"] 中选择答案。你应当给出简洁且分步的解答。最后, 以 "the answer is [ANSWER]" 的格式得出结论, 其中 [ANSWER] 是选项 ["A"、"B"、"C"、"D"] 中的一个。例如, "the answer is A"、"the answer is B"、"the answer is C" 或 "the answer is D"。如果答案不在选项中, 则声明为 Unanswerable。

示例 1

问题: Multiplan Center 的地址是什么?

选项: (A) 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh (B) 38/1/C BC DAS Street Lalbagh (C) Polashi,BUET (D) Central Road, USA

元数据: { "skill": "Fetch context from corresponding google map api and based on the context answer the question" }

Google Maps 响应:

名称: Multiplan Center

地址: 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh

评分: 4.4

类型: point_of_interest, establishment

当前是否营业: False

工作日营业时间:

- 星期一: 10:00AM-8:00PM

- 星期二: Closed

- 星期三: 10:00AM-8:00PM

- 星期四: 10:00AM-8:00PM

- 星期五: 10:00AM-8:00PM

- 星期六: 10:00AM-8:00PM

- 星期日: 10:00AM-8:00PM

解答: 如果你查看上下文并搜索, 那么在到达之后, Multiplan Center 的地址是 69, 71 New Elephant Rd, Dhaka 1205, Bangladesh。因此, the answer is B。

.... <另外 4 个示例>

现在回答以下问题。

问题: {question}

B.3 视觉地点识别器

视觉地点识别器的提示词

我将为你提供一张图像。你必须确定图像内的精确中心位置, 并通过从中心预测一个半径来预测图像的边界。你的返回格式应为: 中心位置名称后跟一个空格, 然后是半径。在此基础上, 仅提供一个完整地址, 采用单一、一致的格式, 后跟一个空格和一个半径。不要在地址之前或之后包含任何多余文本。

C 详细工具参数

以下部分概述了 Map Service 模块所使用的工具, 这些工具由 map-tool agent 检索得到。每个工具都与其输入参数一起定义。

Trip Tool

current_location (str): 行程的起始位置。

visiting_places (list): 要访问的位置列表。

travel_mode (str): 出行方式, 默认为 driving。

Route Tool

origin (str) : 路线的起始位置。
destination (str) : 路线的目的地。
travel_mode (str) : 出行方式，默认为 driving。
mode (str) : 用于路线的交通方式，例如 "driving"
alternatives (bool) : 是否返回多条可能路线。

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Nearby Tool

query (str): The search term to look for geospatial places.

location (str): The name of the current location (e.g., Ibn Sina Hospital, Dhaka).

type (str): The type of place to search for, such as "restaurant", "cafe", or "hospital".

radius (float): The radius of the circular area for filtering, defaults to 20 Kilometer

PlaceInfo Tool

location_address (str): The address or name of the location to search for.

D Detailed Cost Analysis

In this section, we provide a granular breakdown of the cost analysis presented in Section 6. We compare MapAgent against baselines across four datasets (MapEval-API, MapEval-Textual, MapEval-Visual, and MapQA) and four LLM backbones (GPT-3.5-Turbo, Qwen-2.5-72B, GPT-4o, and Qwen-2.5-VL-72B).

The detailed results are presented in the following tables:

- **Table 12:** Average LLM API calls per query.
- **Table 13:** Average token consumption (in thousands).
- **Table 14:** Average Google Maps API calls per query.
- **Table 15:** Average execution time (in seconds).

Consistent with the aggregate trends, MapAgent demonstrates superior efficiency in every subcategory, validating the scalability of the hierarchical parallel architecture.

E Qualitative Examples

E.1 How MapAgent Answer a Query

Listing 1 and Figure 6 illustrates how a MapAgent addresses a typical geospatial query. In this example, the query is taken from the trip category: "I am currently at Planetshakers Church Melbourne

City Campus. I want to drive to Carrot Tattoo Studio and then to Melbourne Lifestyle Apartments - Docklands Accommodation. How long will this take?"

Initially, the query is passed to the Planner Agent, which analyzes it and selects three modules to solve it: google_maps, solution_generator, and answer_generator. Subsequently, the google_maps module is executed and attempts to fetch information from Google Maps by engaging its map-tool agent and relevant tools. The execution is then handed over to the map-tool agent. Based on the query, the map-tool agent selects the trip tool to retrieve the necessary information for this specific question. The trip tool utilizes the Directions and Place Details APIs to fetch information for solving this particular query. The map-tool agent then returns this information to the google_maps module. The Google Maps module then passes this information to the solution_generator. Based on the information fetched by google_maps and the query itself, the solution_generator attempts to formulate the answer with a proper explanation and hands this over to the answer_generator. Finally, after verifying the answer, the answer_generator extracts the final answer from the explanation provided by the solution_generator and presents the final output. By decomposing the task into multiple steps, this approach addresses the long-context processing limitations of LLMs (Parvez, 2025).

E.2 How MapAgent Answer a Query with Visual Context

Listings 2, 3, 4, 5 and Figure 7 illustrate examples of how MapAgent handles geospatial queries that include a visual context, such as a snapshot of a location. To address this type of problem, the question and image are processed by the Planner Agent. The Planner Agent typically selects the Visual Place Recognizer Module, the google_maps module, the solution_generator module, and the answer_generator module. The Image Search module identifies the central location within the image and estimates its boundaries, often providing a radius from the central location. This location information, along with the original query, is then passed to the google_maps module. Google Maps, through its map-tool agent and tools that invoke the Google Maps API, fetches further relevant information from its inner layer. Finally, similar to the previous example, the solution is generated and refined.

Nearby Tool

query (str) : 要查找的地理空间地点的搜索词。
location (str) : 当前地点的名称 (例如 , Ibn Sina Hospital, Dhaka) 。
type (str) : 要搜索的地点类型 , 例如 "restaurant"、"cafe" 或 "hospital"。
radius (float) : 用于过滤的圆形区域半径 , 默认为 20 公里

PlaceInfo Tool

location_address (str) : 要搜索的位置的地址或名称。

D 详细成本分析

在本节中 , 我们提供第 6 节中所述成本分析的细粒度分解。我们将 MapAgent 与基线在四个数据集 (MapEval-API、MapEval-Textual、MapEval-Visual 和 MapQA) 以及四个 LLM 主干 (GPT-3.5-Turbo、Qwen-2.5-72B、GPT-4o 和 Qwen-2.5-VL-72B) 上进行比较。详细结果呈现在以下表格中 :

- 表 12 : 每次查询的平均 LLM API 调用次数。
 - 表 13 : 平均 token 消耗量 (以千为单位) 。
 - 表 14 : 每次查询的平均 Google Maps API 调用次数。
 - 表 15 : 平均执行时间 (以秒为单位) 。
- 与总体趋势一致 , MapAgent 在每个子类别中都展现出卓越的效率 , 验证了分层并行架构的可扩展性。

E 定性示例

E.1 MapAgent 如何回答查询

列表 1 和图 6 展示了 MapAgent 如何处理一个典型的地理空间查询。在此示例中 , 查询取自 trip 类别 : "I am currently at Planetshakers Church Melbourne City Campus. I want to drive to Carrot Tattoo Studio and then to Melbourne Lifestyle Apartments - Docklands Accommodation. How long will this take?"

最初 , 查询被传递给 Planner Agent , 它分析查询并选择三个模块来解决它 : google_maps、solution_generator 和 answer_generator。随后 , google_maps 模块被执行 , 并尝试通过调用其 map-tool agent 和相关工具从 Google Maps 获取信息。执行随后被移交给 map-tool agent。基于查询 , map-tool agent 选择 trip 工具来检索此特定问题所需的信息。trip 工具利用 Directions 和 Place Details API 来获取解决此特定查询的信息。然后 map-tool agent 将此信息返回给 google_maps 模块。Google Maps 模块随后将此信息传递给 solution_generator。基于 google_maps 获取的信息和查询本身 , solution_generator 尝试用适当的解释来制定答案 , 并将其移交给 answer_generator。最后 , 在验证答案后 , answer_generator 从 solution_generator 提供的解释中提取最终答案 , 并呈现最终输出。通过将任务分解为多个步骤 , 这种方法解决了 LLM 的长上下文处理限制 (Parvez, 2025) 。

E.2 MapAgent 如何回答带有视觉上下文的查询

列表 2、3、4、5 和图 7 展示了 MapAgent 如何处理包含视觉上下文 (例如位置的快照) 的地理空间查询的示例。为了解决这类问题 , 问题和图像由 Planner Agent 处理。Planner Agent 通常选择 Visual Place Recognizer Module、google_maps 模块、solution_generator 模块和 answer_generator 模块。Image Search 模块识别图像中的中心位置并估计其边界 , 通常提供距中心位置的半径。此位置信息连同原始查询一起被传递给 google_maps 模块。Google Maps 通过其 map-tool agent 和调用 Google Maps API 的工具 , 从其内层获取进一步的相关信息。最后 , 与前一个示例类似 , 解决方案被生成并精炼。

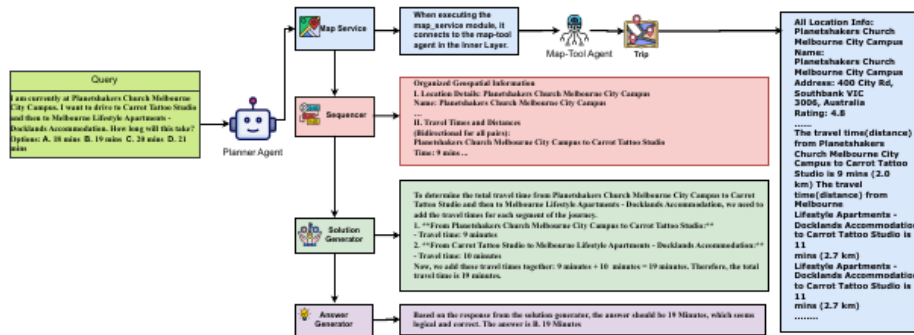


Figure 6: Detailed Example: How MapAgent Solves a Query, Including the Output of Each Module

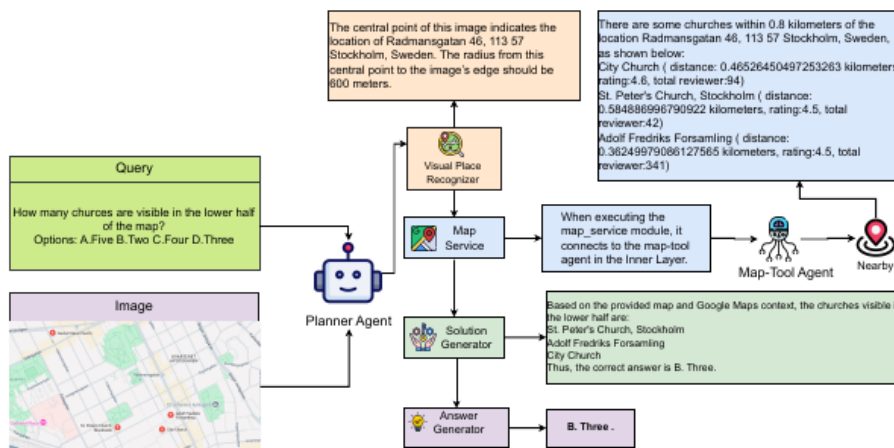


Figure 7: Detailed Example: How MapAgent Solves a Query, Including the Output of Each Module

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解决方案生成器 (Solution Generator)
答案生成器 (Answer Generator)
地图服务 (Map Service)
规划智能体 (Planner Agent)
地图工具智能体 (Map-Tool Agent)
在执行 map_service 模块时，它会连接到内层 (Inner Layer) 的 map-tool agent。
所有位置信息：

Planetshakers Church Melbourne City Campus
名称：
Planetshakers Church Melbourne City Campus
地址：400 City Rd, Southbank VIC 3006, Australia
评分：4.8

.....
从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio 的出行时间 (距离) 为 9 分钟 (2.0 km)。从 Melbourne Lifestyle Apartments - Docklands Accommodation 到 Carrot Tattoo Studio 的出行时间 (距离) 为 11 分钟 (2.7 km)。
Lifestyle Apartments - Docklands Accommodation 到 Carrot Tattoo Studio 为 11 分钟 (2.7 km)
.....

为了确定从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio，然后到 Melbourne Lifestyle Apartments - Docklands Accommodation 的总出行时间，我们需要将旅程每一段的出行时间相加。
1. 从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio：
- 出行时间：9 分钟
2. 从 Carrot Tattoo Studio 到 Melbourne Lifestyle Apartments - Docklands Accommodation：
- 出行时间：10 分钟
现在，我们将这些出行时间相加：9 分钟 + 10 分钟 = 19 分钟。因此，总出行时间为 19 分钟。
根据 solution generator 的响应，答案应为 19 Minutes，这看起来合理且正确。答案是 B. 19 Minutes
我目前在 Planetshakers Church Melbourne City Campus。我想开车去 Carrot Tattoo Studio，然后去 Melbourne Lifestyle Apartments - Docklands Accommodation。这会花多长时间？
选项：A. 18 mins B. 19 mins C. 20 mins D. 21 mins

查询 (Query)
序列器 (Sequencer)
行程 (Trip)
组织后的地理空间信息 (Organized Geospatial Information)
I. 位置详情：Planetshakers Church Melbourne City Campus
名称：Planetshakers Church Melbourne City Campus
...
II. 出行时间与距离
(所有配对均为双向)：
Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio
时间：9 mins ...

图 6：详细示例：MapAgent 如何解决一个查询，包括每个模块的输出
解决方案生成器 (Solution Generator)

答案生成器 (Answer Generator)

视觉地点识别器 (Visual Place Recognizer)

地图服务 (Map Service)

规划智能体 (Planner Agent)

地图工具智能体 (Map-Tool Agent)

附近 (Nearby)

这张图像的中心点表示 Radmansgatan 46, 113 57 Stockholm, Sweden 的位置。从该中心点到图像边缘的半径应为 600 米。

在执行 map_service 模块时，它会连接到内层 (Inner Layer) 的 map-tool agent。

在位置 Radmansgatan 46, 113 57 Stockholm, Sweden 的 0.8 公里范围内有一些教堂，如下所示：

City Church (距离：0.46526450497253263 公里，评分：4.6，总评论数：94)

St. Peter's Church, Stockholm (距离：0.584886996790922 公里，评分：4.5，总评论数：42)

Adolf Fredriks Forsamling (距离：0.36249979086127565 公里，评分：4.5，总评论数：341)

根据提供的地图和 Google Maps 上下文，在地图下半部分可见的教堂是：

St. Peter's Church, Stockholm

Adolf Fredriks Forsamling

City Church

因此，正确答案是 B. Three。

B. Three。

地图下半部分可见多少座教堂？

选项：A.Five B.Two C.Four D.Three

查询 (Query)

图像 (Image)

图 7：详细示例：MapAgent 如何解决一个查询，包括每个模块的输出

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Table 9: Detailed Reasoning Steps of Octotools

Step	Tool	Sub-goal	Command	Result Summary
Action Step 1	Route Tool	Calculate the total driving time for the round trip starting at Rydges Wellington, visiting Kelburn Park and Victoria University of Wellington, and returning to Rydges Wellington.	<code>execution=tool.execute(origin="RydgesWellington",destination="RydgesWellington",waypoints=["KelburnPark","VictoriaUniversityofWellington"],mode="driving",alternatives=False)</code>	Total 1 route. VIA Dixon St (1.9 km, 9 mins). Details include step-by-step directions.
Action Step 2	Route Tool	Calculate the travel time from the Trevi Fountain to the Vatican Museum using public transport.	<code>execution=tool.execute(origin="TreviFountain",destination="VaticanMuseum",mode="transit")</code>	Total 6 routes from Trevi Fountain to Vatican Museum via different public transport options with times ranging from 30 to 39 mins.
Action Step 3	Route Tool	Calculate the travel time for the transit leg from Kockelscheuer Ice Rink to Kallektuffquell using the Route_Tool.	<code>execution=tool.execute(origin="KockelscheuerIceRink",destination="Kallektuffquell",mode="transit")</code>	Total 6 routes from Kockelscheuer Ice Rink to Kallektuffquell via public transport with times ranging from 1 hour 50 mins to 2 hours 10 mins.
Action Step 4	Route Tool	Calculate the total driving time for the route from Planetshakers Church Melbourne City Campus to Melbourne Lifestyle Apartments - Docklands Accommodation with a stop at Carrot Tattoo Studio	<code>execution=tool.execute(origin="PlanetshakersChurchMelbourneCityCampus",destination="MelbourneLifestyleApartments-DocklandsAccommodation",waypoints=["CarrotTattooStudio"],mode="driving",alternatives=False)</code>	Total 1 route from Planetshakers Church to Melbourne Lifestyle Apartments (via Carrot Tattoo Studio) (2.0 km, 9 mins) with step-by-step directions.
Action Steps 5-10	Route Tool	Calculate the total driving time for the route from Hotel Saltel to Nil Ratan Sircar Medical College and Hospital with a stop at Science City - Kolkata.	<code>execution=tool.execute(origin="HotelSaltel",destination="NilRatanSircarMedicalCollegeandHospital",waypoints=["ScienceCity-Kolkata"],mode="driving",alternatives=False)</code>	Each step shows 1 route from Hotel Saltel to Nil Ratan Sircar Medical College (via Science City - Kolkata) (2.8 km, 7 mins) with step-by-step directions.

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表 9 : Octotools 的详细推理步骤

步骤	工具	子目标	命令	结果摘要
动作				
步骤 1				
Route				
Tool				
				计算从 Rydges Wellington 出发，途经 Kelburn Park 和 Victoria University of Wellington，并返回 Rydges Wellington 的往返总驾车时间。
				<code>execution=tool.execute(origin="RydgesWellington",destination="RydgesWellington",waypoints=["KelburnPark","VictoriaUniversityofWellington"],mode="driving",alternatives=False)</code>
				共 1 条路线。经由 Dixon St (1.9 km，9 分钟)。详情包括逐步导航指引。
动作				
步骤 2				
Route				
Tool				
				计算从 Trevi Fountain 到 Vatican Museum 使用公共交通的出行时间。
				<code>execution=tool.execute(origin="TreviFountain",destination="VaticanMuseum",mode="transit")</code>
				从 Trevi Fountain 到 Vatican Museum 共有 6 条路线，经由不同公共交通选项，时间范围为 30 至 39 分钟。
动作				
步骤 3				
Route				
Tool				
				使用 Route_Tool 计算从 Kockelscheuer Ice Rink 到 Kallektuffquell 的公共交通路段出行时间。
				<code>execution=tool.execute(origin="KockelscheuerIceRink",destination="Kallektuffquell",mode="transit")</code>
				从 Kockelscheuer Ice Rink 到 Kallektuffquell 共有 6 条公共交通路线，时间范围为 1 小时 50 分钟至 2 小时 10 分钟。
动作				
步骤 4				
Route				
Tool				
				计算从 Planetshakers Church Melbourne City Campus 到 Melbourne Lifestyle Apartments - Docklands Accommodation 并在 Carrot Tattoo Studio 停靠的路线总驾车时间
				<code>execution=tool.execute(origin="PlanetshakersChurchMelbourneCityCampus",destination="MelbourneLifestyleApartments-DocklandsAccommodation",waypoints=["CarrotTattooStudio"],mode="driving",alternatives=False)</code>
				从 Planetshakers Church 到 Melbourne Lifestyle Apartments (经由 Carrot Tattoo Studio) 共 1 条路线 (2.0 km，9 分钟)，包含逐步导航指引。
动作				
步骤				
5-10				
Route				
Tool				
				计算从 Hotel Saltel 到 Nil Ratan Sircar Medical College and Hospital 并在 Science City - Kolkata 停靠的路线总驾车时间。
				<code>execution=tool.execute(origin="HotelSaltel",destination="NilRatanSircarMedicalCollegeandHospital",waypoints=["Sc</code>

ienceCity-Kolkata"],mode="driving",alternatives=False)

每一步显示从 Hotel Saltel 到 Nil Ratan Sircar Medical College (经由 Science City - Kolkata) 的 1 条路线 (2.8 km , 7 分钟) , 包含逐步导航指引。

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Module	Description
Map Service	Fetches geospatial data using a Coordinator Agent called Map-Tool Agent. Retrieves place details, routes, nearby locations, and other map-related information.
Visual Place Recognizer	Uses VLLM to analyze geospatial imagery. Identifies the central geographic place and estimates the geographic scope (approximate radius).
Sequencer	Organizes and structures the responses from preceding modules into a logical sequence.
Solution Generator	Uses a LLM or VLLM to synthesize the final answer based on structured Input.
Answer Generator	Refines and verifies the solution produced by the Solution Generator. Extracts and checks the predicted answer for consistency.

Table 10: Modules in the Module Inventory essential for solving geospatial queries

Tool	Description
Trip Tool	Retrieves information about locations and travel routes between them. Combines the Place Details and Directions APIs to fetch place metadata and step-by-step route instructions.
Route Tool	Fetches route information between two points using the Directions API, including distance, estimated time, and navigation steps.
Nearby Tool	Identifies points of interest within a specified area using the Nearby Search API. Returns place names, ratings, and other metadata.
PlaceInfo Tool	Retrieves detailed information about a given place using the Place Details API, including address, hours, contact info, and reviews.

Table 11: Map tools used by the MapAgent framework, each built over a specific Google Maps API.

LLM Backbone	Dataset	Chameleon	Octotools	MapAgent (Ours)
GPT-3.5-Turbo	MapEval-API	7.0	12.0	6.0
	MapEval-Textual	6.5	11.0	5.5
	MapQA	6.0	10.0	5.0
Qwen-2.5-72B	MapEval-API	7.0	12.0	6.0
	MapEval-Textual	6.5	11.0	5.5
	MapQA	6.0	10.0	5.0
GPT-4o	MapEval-Visual	8.0	14.0	7.0
Qwen-2.5-VL-72B	MapEval-Visual	8.0	14.0	7.0

Table 12: Comparison of average LLM API calls per query across different datasets and backbones. MapAgent consistently demonstrates higher efficiency by requiring fewer LLM API interactions than baselines.

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模块 描述

Map Service 使用名为 Map-Tool Agent 的 Coordinator Agent 获取地理空间数据。

检索地点详情、路线、附近位置及其他地图相关信息。

Visual Place

Recognizer 使用 VLLM 分析地理空间影像。识别中心地理地点并估计地理范围（近似半径）。

Sequencer 将前述模块的响应组织并结构化为逻辑序列。

Solution Generator 使用 LLM 或 VLLM 基于结构化输入合成最终答案。

Answer Generator 对 Solution Generator 产生的解决方案进行精炼和验证。提取并检查预测答案的一致性。

表 10：Module Inventory 中对于解决地理空间查询必不可少的模块

工具 描述

Trip Tool 检索位置信息及它们之间的旅行路线。结合 Place Details 和 Directions API 获取地点元数据和逐步路线指引。

Route Tool 使用 Directions API 获取两点之间的路线信息，包括距离、预计时间和导航步骤。

Nearby Tool 使用 Nearby Search API 识别指定区域内的兴趣点。返回地点名称、评分及其他元数据。

PlaceInfo Tool 使用 Place Details API 检索给定地点的详细信息，包括地址、营业时间、联系信息和评论。

表 11：MapAgent 框架使用的地图工具，每个工具都构建在特定的 Google Maps API 之上。

LLM Backbone 数据集 Chameleon Octotools MapAgent (Ours)

GPT-3.5-Turbo

MapEval-API 7.0 12.0 6.0

MapEval-Textual 6.5 11.0 5.5

MapQA 6.0 10.0 5.0

Qwen-2.5-72B

MapEval-API 7.0 12.0 6.0

MapEval-Textual 6.5 11.0 5.5

MapQA 6.0 10.0 5.0

GPT-4o MapEval-Visual 8.0 14.0 7.0

Qwen-2.5-VL-72B MapEval-Visual 8.0 14.0 7.0

表 12：不同数据集和骨干模型下每次查询的平均 LLM API 调用次数比较。MapAgent 始终表现出更高的效率，所需 LLM API 交互次数少于基线。

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LLM Backbone	Dataset	Chameleon	Octotools	MapAgent (Ours)
GPT-3.5-Turbo	MapEval-API	4.5	8.0	3.0
	MapEval-Textual	5.0	10.0	3.5
	MapQA	4.0	7.0	2.5
Qwen-2.5-72B	MapEval-API	4.5	8.0	3.0
	MapEval-Textual	5.0	10.0	3.5
	MapQA	4.0	7.0	2.5
GPT-4o	MapEval-Visual	6.0	10.0	4.0
Qwen-2.5-VL-72B	MapEval-Visual	6.0	10.0	4.0

Table 13: Comparison of average token consumption (in thousands, K) per query. MapAgent significantly reduces token usage compared to baselines, indicating more efficient context management.

LLM Backbone	Dataset	Chameleon	Octotools	MapAgent (Ours)
GPT-3.5-Turbo	MapEval-API	2.0	6.0	2.0
	MapEval-Textual	1.5	4.0	1.5
	MapQA	1.0	3.0	1.0
Qwen-2.5-72B	MapEval-API	2.0	6.0	2.0
	MapEval-Textual	1.5	4.0	1.5
	MapQA	1.0	3.0	1.0
GPT-4o	MapEval-Visual	2.0	5.0	2.0
Qwen-2.5-VL-72B	MapEval-Visual	2.0	5.0	2.0

Table 14: Comparison of average Google Maps API calls per query. MapAgent minimizes tool usage, achieving parity or improvement over baselines.

LLM Backbone	Dataset	Chameleon	Octotools	MapAgent (Ours)
GPT-3.5-Turbo	MapEval-API	23.0	38.0	15.0
	MapEval-Textual	20.0	34.0	13.0
	MapQA	17.0	28.0	14.0
Qwen-2.5-72B	MapEval-API	29.0	42.0	20.0
	MapEval-Textual	24.0	41.0	16.0
	MapQA	22.0	36.0	15.0
GPT-4o	MapEval-Visual	25.0	40.0	17.0
Qwen-2.5-VL-72B	MapEval-Visual	32.0	46.0	22.0

Table 15: Comparison of average execution time (in seconds) per query. MapAgent achieves significantly lower latency across all benchmarks compared to the baselines.

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LLM Backbone Dataset Chameleon Octotools MapAgent (Ours)
GPT-3.5-Turbo
MapEval-API 4.5 8.0 3.0
MapEval-Textual 5.0 10.0 3.5
MapQA 4.0 7.0 2.5
Qwen-2.5-72B
MapEval-API 4.5 8.0 3.0
MapEval-Textual 5.0 10.0 3.5
MapQA 4.0 7.0 2.5
GPT-4o MapEval-Visual 6.0 10.0 4.0

Qwen-2.5-VL-72B MapEval-Visual 6.0 10.0 4.0

表 13：每次查询平均 token 消耗量（以千为单位，K）的比较。与基线相比，MapAgent 显著降低了 token 使用量，表明其上下文管理更加高效。

LLM Backbone Dataset Chameleon Octotools MapAgent (Ours)
GPT-3.5-Turbo
MapEval-API 2.0 6.0 2.0
MapEval-Textual 1.5 4.0 1.5
MapQA 1.0 3.0 1.0
Qwen-2.5-72B
MapEval-API 2.0 6.0 2.0
MapEval-Textual 1.5 4.0 1.5
MapQA 1.0 3.0 1.0
GPT-4o MapEval-Visual 2.0 5.0 2.0

Qwen-2.5-VL-72B MapEval-Visual 2.0 5.0 2.0

表 14：每次查询平均 Google Maps API 调用次数的比较。MapAgent 将工具使用降至最低，达到与基线持平或更优的水平。

LLM Backbone Dataset Chameleon Octotools MapAgent (Ours)
GPT-3.5-Turbo
MapEval-API 23.0 38.0 15.0
MapEval-Textual 20.0 34.0 13.0
MapQA 17.0 28.0 14.0
Qwen-2.5-72B
MapEval-API 29.0 42.0 20.0
MapEval-Textual 24.0 41.0 16.0
MapQA 22.0 36.0 15.0
GPT-4o MapEval-Visual 25.0 40.0 17.0

Qwen-2.5-VL-72B MapEval-Visual 32.0 46.0 22.0

表 15：每次查询平均执行时间（以秒为单位）的比较。与基线相比，MapAgent 在所有基准测试上均实现了显著更低的延迟。

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Listing 1: Example how MapAgent answer a query **Green**: Correct Answer. **Red**: Wrong Answer.

```
Question:
I am currently at Planetshakers Church Melbourne City Campus. I want to drive to Carrot Tattoo Studio and
then to Melbourne Lifestyle Apartments - Docklands Accommodation. How long will this take?
Options: A. 18 mins B. 19 mins C. 20 mins D. 21 mins

Modules selected by Planner agent:
1.Map Service
2.Solution Generator
3.Answer Generator

Information gathered by Map Service module via Map-Tool Agent (invoking Trip tool):
All Location Info:
Planetshakers Church Melbourne City Campus
Name: Planetshakers Church Melbourne City Campus
Address: 400 City Rd, Southbank VIC 3006, Australia
Rating: 4.8
Types: church, tourist_attraction, place_of_worship, point_of_interest, establishment
Is Open Now: False
Weekday Opening Hours:
- Monday: Closed
....
- Saturday: Closed
- Sunday: 9:00AM - 8:00PM
Carrot Tattoo Studio
Name: Carrot Tattoo Studio
Address: Basement/63 Elizabeth St, Melbourne VIC 3000, Australia
Rating: 5
Types: store, point_of_interest, establishment
Is Open Now: False
Weekday Opening Hours:
- Monday: 10:00 AM - 8:00 PM
- Tuesday: 10:00AM - 8:00PM
....
- Sunday: 10:00AM - 8:00PM
Melbourne Lifestyle Apartments - Docklands Accommodation
Name: Melbourne Lifestyle Apartments - Best Views on Collins
Address: 893A Collins St, Docklands VIC 3008, Australia
Rating: 4.8
Types: lodging, point_of_interest, establishment
Is Open Now: True
Weekday Opening Hours:
- Monday: Open 24 hours
- Tuesday: Open 24 hours
....
The travel time(distance) from Planetshakers Church Melbourne City Campus to Carrot Tattoo Studio is 9 mins
(2.0 km)
The travel time(distance) from Planetshakers Church Melbourne City Campus to Melbourne Lifestyle Apartments
- Docklands Accommodation is 8 mins (2.5 km)
The travel time(distance) from Carrot Tattoo Studio to Planetshakers Church Melbourne City Campus is 9 mins
(1.9 km)
....
The travel time(distance) from Melbourne Lifestyle Apartments - Docklands Accommodation to Carrot Tattoo
Studio is 11 mins (2.7 km)

Solution generated by solution_generator module:
To determine the total travel time from Planetshakers Church Melbourne City Campus to Carrot Tattoo Studio
and then to Melbourne Lifestyle Apartments - Docklands Accommodation, we need to add the travel times for
each segment of the journey.

1. **From Planetshakers Church Melbourne City Campus to Carrot Tattoo Studio:**
   - Travel time: 9 minutes

2. **From Carrot Tattoo Studio to Melbourne Lifestyle Apartments - Docklands Accommodation:**
   - Travel time: 10 minutes

Now, we add these travel times together:
9 minutes + 10 minutes = 19 minutes

Therefore, the total travel time is 19 minutes.

Correct option extracted by answer_generator module: B. 19 minutes
```

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Listing 1 : MapAgent 回答查询的示例 绿色 : 正确答案。红色 : 错误答案。

问题 :

我目前在 Planetshakers Church Melbourne City Campus。我想开车去 Carrot Tattoo Studio , 然后去 Melbourne Lifestyle Apartments - Docklands Accommodation。这会花多长时间 ?

选项 : A. 18 mins B. 19 mins C. 20 mins D. 21 mins

Planner agent 选择的模块 :

1. Map Service
2. Solution Generator
3. Answer Generator

Map Service 模块通过 Map-Tool Agent (调用 Trip tool) 收集的信息 :

所有位置信息 :

Planetshakers Church Melbourne City Campus

名称 : Planetshakers Church Melbourne City Campus

地址 : 400 City Rd, Southbank VIC 3006, Australia

评分 : 4.8

类型 : church, tourist_attraction, place_of_worship, point_of_interest, establishment

当前是否营业 : False

工作日营业时间 :

- Monday : Closed

....

....

- Saturday : Closed

- Sunday : 9:00 AM - 8:00 PM

Carrot Tattoo Studio

名称 : Carrot Tattoo Studio

地址 : Basement /63 Elizabeth St, Melbourne VIC 3000, Australia

评分 : 5

类型 : store, point_of_interest, establishment

当前是否营业 : False

工作日营业时间 :

- Monday : 10:00 AM - 8:00 PM

- Tuesday : 10:00 AM - 8:00 PM

....

....

- Sunday : 10:00 AM - 8:00 PM

Melbourne Lifestyle Apartments - Docklands Accommodation

名称 : Melbourne Lifestyle Apartments - Best Views on Collins

地址 : 893 A Collins St, Docklands VIC 3008, Australia

评分 : 4.8

类型 : lodging, point_of_interest, establishment

当前是否营业 : True

工作日营业时间：

- Monday : Open 24 hours
- Tuesday : Open 24 hours

....

....

从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio 的行程时间 (距离) 为 9 mins (2.0 km)

从 Planetshakers Church Melbourne City Campus 到 Melbourne Lifestyle Apartments

- Docklands Accommodation 的行程时间 (距离) 为 8 mins (2.5 km)

从 Carrot Tattoo Studio 到 Planetshakers Church Melbourne City Campus 的行程时间 (距离) 为 9 mins (1.9 km)

....

....

从 Melbourne Lifestyle Apartments - Docklands Accommodation 到 Carrot Tattoo Studio 的行程时间 (距离) 为 11 mins (2.7 km)

solution_generator 模块生成的解答：

为了确定从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio , 然后到 Melbourne Lifestyle Apartments - Docklands Accommodation 的总行程时间，我们需要将每一段行程的行程时间相加。

1. 从 Planetshakers Church Melbourne City Campus 到 Carrot Tattoo Studio：

- 行程时间：9 minutes

2. 从 Carrot Tattoo Studio 到 Melbourne Lifestyle Apartments - Docklands Accommodation：

- 行程时间：10 minutes

现在，我们将这些行程时间相加：

9 minutes + 10 minutes = 19 minutes

因此，总行程时间是 19 minutes。

answer_generator 模块提取出的正确选项：B.19 minutes

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Listing 2: Example how MapAgent answer MapEval-Visual counting query **Green**: Correct Answer. **Red**: Wrong Answer.

Context:



Question:

How many churches are visible in the lower half of the map?

Options: A. Five B. Two C. Four **D. Three**

Modules selected by Planner agent:

Module Selection from the Module Catalog and their execution order:

1. Visual Place Recognizer module
2. map_service module
3. solution_generator module
4. answer_generator module

Visual Place Recognizer module Output:

This image center represents the location of Rådmansgatan 46, 113 57 Stockholm, Sweden.

Map Service module Output:

When executing the map_service module, it connects to the map-tool agent in the Inner Layer.

Map-Tool Agent Output:

Nearby tool is selected to fetch information.

Nearby Tool's Output:

There are some churches within 0.8 kilometers of the location Rådmansgatan 46, 113 57 Stockholm, Sweden, as

shown below:

City Church (distance: 0.46526450497253263 kilometers, rating:4.6, total reviewer:94)

St. Peter's Church, Stockholm (distance: 0.584886996790922 kilometers, rating:4.5, total reviewer:42)

Adolf Fredriks församling (distance: 0.36249979086127565 kilometers, rating:4.5, total reviewer:341)

solution generator module's Output:

Based on the provided map and Google Maps context, the churches visible in the lower half are:

St. Peter's Church, Stockholm

Adolf Fredriks församling

City Church

Thus, the correct answer is B. Three.

answer generator module's Output:

Based on the response from the solution generator, the answer should be 3, which seems logical and correct.

The answer is **B. Three**.

Comments:

The Final answer B. 3 is a correct Answer

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Listing 2: MapAgent 如何回答 MapEval-Visual 计数查询的示例 绿色：正确答案。红色：错误答案。

上下文：

问题：

地图下半部分可以看到多少座教堂？

选项：A. 五座 B. 两座 C. 四座 D. 三座

Planner agent 选择的模块：

从模块目录中选择的模块及其执行顺序：

1. Visual Place Recognizer 模块

2. map_service 模块

3. solution_generator 模块

4. answer_generator 模块

Visual Place Recognizer 模块输出：

该图像中心表示瑞典斯德哥尔摩 Radmansgatan 46, 113 57 的位置。

Map Service 模块输出：

执行 map_service 模块时，它会连接到内层（Inner Layer）中的 map-tool agent。

Map-Tool Agent 输出：

选择 Nearby 工具来获取信息。

Nearby 工具的输出：

在瑞典斯德哥尔摩 Radmansgatan 46, 113 57 位置 0.8 公里范围内有一些教堂，如下所示：

City Church (距离：0.46526450497253263 公里，评分：4.6，评论者总数：94)

St. Peter's Church, Stockholm (距离：0.584886996790922 公里，评分：4.5，评论者总数：42)

Adolf Fredriks Forsamling (距离：0.36249979086127565 公里，评分：4.5，评论者总数：341)

solution generator 模块的输出：

根据提供的地图和 Google Maps 上下文，下半部分可见的教堂是：

St. Peter's Church, Stockholm

Adolf Fredriks Forsamling

City Church

因此，正确答案是 B. 三座。

answer generator 模块的输出：

根据 solution generator 的响应，答案应为 3，这看起来合乎逻辑且正确。

答案是 B. 三座。

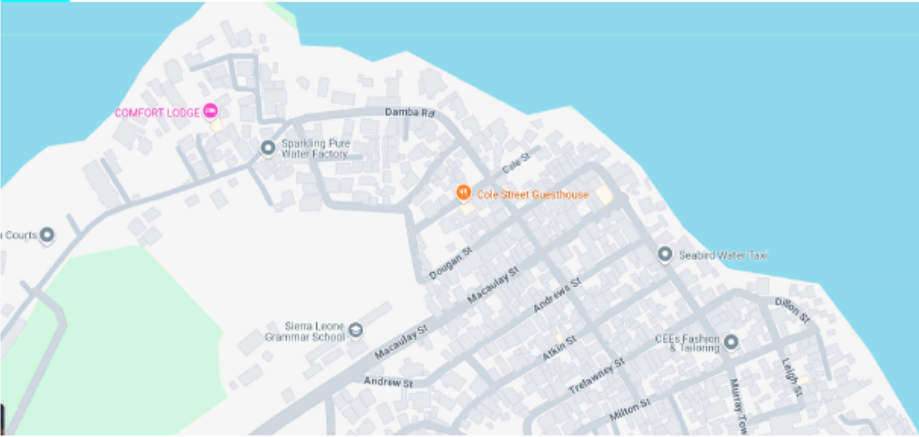
评论：

最终答案 B. 3 是正确答案

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Listing 3: Example how MapAgent answer MapEval-Visual Route query **Green**: Correct Answer. **Red**: Wrong Answer.

Context:



Question:

How to get to Sierra Leone Grammer School from Seabird Water Taxi after having lunch at Cole Street Guesthouse?

Options:

A. Seabird Water Taxi -> Head northwest on High Broad St toward Andrews St -> Turn left onto Andrews St -> Continue onto Andrew St -> Turn right onto Macaulay St -> Sierra Leone Grammar School -> Head northeast on Macaulay St -> Turn left at the 1st cross street toward Cole St -> Turn left at the 2nd cross street onto Cole St -> Cole Street Guesthouse

B. Seabird Water Taxi -> Head northwest on High Broad St toward Andrews St -> Turn left onto Andrews St -> Continue onto Andrew St -> Turn right onto Macaulay St -> Sierra Leone Grammar School

C. Seabird Water Taxi -> Head northwest on High Broad St toward Andrews St -> Turn left onto Macaulay St -> Sierra Leone Grammar School -> Head northeast on Macaulay St -> Turn left at the 1st cross street toward Cole St -> Turn left at the 2nd cross street onto Cole St -> Cole Street Guesthouse

D. Seabird Water Taxi -> Head northwest on High Broad St toward Andrews St -> Turn right toward Cole St -> Turn left at the 1st cross street onto Cole St -> Cole Street Guesthouse -> Head northeast on Cole St toward Damba Rd -> Turn right at the 1st cross street toward Macaulay St -> Turn right at the 2nd cross street onto Macaulay St -> Sierra Leone Grammar School

Modules selected by Planner agent:

Module Selection from the Module Catalog and their execution order:

1. Visual Place Recognizer module

2. map_service module

3. solution_generator module

4. answer_generator module

Image Search module Output:

This image center represents the location of Cole Street Guesthouse, 5 Cole St, Freetown, Sierra Leone.

Map Service module Output:

When executing the map_service module, it connects to the map-tool agent in the Inner Layer.

Map-Tool Agent Output:

Route tool is selected to fetch information.

Route Tool's Output:

There is a total of one route from Seabird Water Taxi, Freetown, Sierra Leone, to Sierra Leone Grammar School, Freetown, Sierra Leone, with a waypoint at Cole Street Guesthouse. The route information is provided below:

Route 1:(VIA) High Broad St (0.4 km, 5 mins)

Details steps are provided below:

Head northwest on High Broad St toward Andrews St (0.3 km, 4 mins)

Turn right toward Cole St (56 m, 1 min)

Turn left onto Cole StDestination will be on the left (46 m, 1 min)

Head northeast on cole st toward Damba Rd (46m, 1 min)

Turn right at the 1st cross street toward Macaulay St (110 m, 1.5 min)

Turn right at the 2nd cross street onto Macaulay St (260 m, 2 min)

Sierra Leone Grammar School

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Listing 3: MapAgent 回答 MapEval-Visual Route 查询的示例 绿色：正确答案。红色：错误答案。

上下文：

问题：

在 Cole Street Guesthouse 吃完午饭后，如何从 Seabird Water Taxi 前往 Sierra Leone Grammar School ?

选项：

A . Seabird Water Taxi -> 沿 High Broad St 向西北方向朝 Andrews St 行进 -> 左转进入 Andrews St ->

继续进入 Andrew St -> 右转进入 Macaulay St -> Sierra Leone Grammar School -> 沿 Macaulay St 向东北方向行进 ->

在第 1 个十字路口左转朝 Cole St 方向 -> 在第 2 个十字路口左转进入

Cole St -> Cole Street Guesthouse

B . Seabird Water Taxi -> 沿 High Broad St 向西北方向朝 Andrews St 行进 -> 左转进入 Andrews St ->

继续进入 Andrew St -> 右转进入 Macaulay St -> Sierra Leone Grammar School

C . Seabird Water Taxi -> 沿 High Broad St 向西北方向朝 Andrews St 行进 -> 左转进入 Macaulay St ->

Sierra Leone Grammar School -> 沿 Macaulay St 向东北方向行进 -> 在第 1 个十字路口左转朝

Cole St 方向 -> 在第 2 个十字路口左转进入 Cole St -> Cole Street Guesthouse

D. Seabird Water Taxi -> 沿 High Broad St 向西北方向朝 Andrews St 行进 -> 右转朝 Cole St 方向 ->

在第 1 个十字路口左转进入 Cole St -> Cole Street Guesthouse -> 沿 Cole St 向东北方向朝

Damba Rd 行进 -> 在第 1 个十字路口右转朝 Macaulay St 方向 -> 在第 2 个十字路口右转

进入 Macaulay St -> Sierra Leone Grammar School

Planner agent 选择的模块：

从模块目录中选择的模块及其执行顺序：

1. Visual Place Recognizer 模块

2. map_service 模块

3. solution_generator 模块

4. answer_generator 模块

Image Search 模块输出：

该图像中心表示 Cole Street Guesthouse，5 Cole St，Freetown，Sierra Leone 的位置。

Map Service 模块输出：

执行 map_service 模块时，它连接到 Inner Layer 中的 map - tool agent。

Map-Tool Agent 输出：

选择 Route tool 来获取信息。

Route Tool 的输出：

从 Seabird Water Taxi，Freetown，Sierra Leone 到 Sierra Leone Grammar

School，Freetown，Sierra Leone 共有 1 条路线，途中经过 Cole Street Guesthouse。路线信息提供如下：

Route 1:(VIA) High Broad St (0.4 km，5 mins)

详细步骤如下：

沿 High Broad St 向西北方向朝 Andrews St 行进 (0.3 km，4 mins)

右转朝 Cole St 方向 (56 m，1 min)

左转进入 Cole St 目的地将在左侧 (46 m，1 min)

沿 cole st 向东北方向朝 Damba Rd 行进 (46 m，1 min)

在第 1 个十字路口右转朝 Macaulay St 方向 (110 m , 1.5 min)

在第 2 个十字路口右转进入 Macaulay St (260 m , 2 min)

Sierra Leone Grammar School

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Listing 4: Continuation of Example how MapAgent answer MapEval-Visual Route query **Green**: Correct Answer.

Red: Wrong Answer.

solution generator module's Output:

Based on the provided map and Google Maps context, the best way point is:
Head northwest on High Broad St toward Andrews St
Turn right toward Cole St
Turn left onto Cole St Destination will be on the left
Head northeast on cole st toward Damba Rd
Turn right at the 1st cross street toward Macaulay St
Turn right at the 2nd cross street onto Macaulay St
Sierra Leone Grammar School

Therefor the correct answer is:

**** (D) Seabird Water Taxi**
Head northwest on High Broad St toward Andrews St
Turn right toward Cole St
Turn left at the 1st cross street onto Cole St
Cole Street Guesthouse
Head northeast on Cole St toward Damba Rd
Turn right at the 1st cross street toward Macaulay St
Turn right at the 2nd cross street onto Macaulay St
Sierra Leone Grammar School ******

answer generator module's Output:

Based on the response from the solution generator, the answer should be D, which seems logical and correct.
The answer is **D**. Seabird Water Taxi -> Head northwest on High Broad St toward Andrews St -> Turn right toward Cole St -> Turn left at the 1st cross street onto Cole St -> Cole Street Guesthouse -> Head northeast on Cole St toward Damba Rd -> Turn right at the 1st cross street toward Macaulay St -> Turn right at the 2nd cross street onto Macaulay St -> Sierra Leone Grammar School

Comments:

The Final answer D is a correct Answer

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Listing 4 : MapAgent 如何回答 MapEval-Visual Route 查询的示例续篇 绿色 : 正确答案。

红色 : 错误答案。

solution generator 模块的输出 :

基于提供的地图和 Google Maps 上下文, 最佳路径点是 :

沿 High Broad St 向西北方向前进, 朝 Andrews St 方向

向右转, 朝 Cole St 方向

向左转进入 Cole St目的地将在左侧

沿 cole st 向东北方向前进, 朝 Damba Rd 方向

在第 1 个十字路口向右转, 朝 Macaulay St 方向

在第 2 个十字路口向右转进入 Macaulay St

Sierra Leone Grammar School

因此正确答案是 :

**(D) Seabird Water Taxi

沿 High Broad St 向西北方向前进, 朝 Andrews St 方向

向右转, 朝 Cole St 方向

在第 1 个十字路口向左转进入 Cole St

Cole Street Guesthouse

沿 Cole St 向东北方向前进, 朝 Damba Rd 方向

在第 1 个十字路口向右转, 朝 Macaulay St 方向

在第 2 个十字路口向右转进入 Macaulay St

Sierra Leone Grammar School **

answer generator 模块的输出 :

基于 solution generator 的响应, 答案应为 D, 这看起来合乎逻辑且正确。

答案是 D。Seabird Water Taxi -> 沿 High Broad St 向西北方向前进, 朝 Andrews St 方向 -> 向右转

朝 Cole St 方向 -> 在第 1 个十字路口向左转进入 Cole St -> Cole Street Guesthouse -> 沿 Cole St 向东北方向

前进, 朝 Damba Rd 方向 -> 在第 1 个十字路口向右转, 朝 Macaulay St 方向 -> 在第 2

个十字路口向右转进入 Macaulay St -> Sierra Leone Grammar School

评论 :

最终答案 D 是正确答案

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Listing 5: Example how MapAgent answer MapEval-Visual Nearby query **Green**: Correct Answer. **Red**: Wrong Answer.

Context:



Question:

Which castle is closest to Howth rail station?

Options:

A. Howth Castle

B. Mary Castle

C. Claremont Castle

D. Baliscadden House

Modules selected by Planner agent:

Module Selection from the Module Catalog and their execution order:
1. visual_place_recognizer module
2. map_service module
3. solution_generator module
4. answer_generator module

Visual Place Recognizer module Output:

The center of the image appears to be Howth, Ireland.

Map Service module Output:

When executing the map_service module, it connects to the map-tool agent in the Inner Layer.

Map-Tool Agent Output:

Nearby tool is selected to fetch information.

Nearby Tool's Output:

There are some castle distance from the current location Howth Market, 3 Harbour Rd, Howth, Dublin, D13 T8Y7, Ireland within 15 Kilometers in below:
Howth Castle Estate (distance: 0.4770576879845442 kilometers, rating:4.1, total reviewer:714)
Malahide Castle & Gardens (distance: 8.742973877916125 kilometers, rating:4.6, total reviewer:15235)
Dublin Castle (distance: 13.875697894024961 kilometers, rating:4.3, total reviewer:33876)
Swords Castle (distance: 12.548387008403804 kilometers, rating:4.5, total reviewer:1764)
Dalkey Castle & Heritage Centre (distance: 12.505523354933828 kilometers, rating:4.6, total reviewer:911)
Monkstown Castle (distance: 12.235452667377206 kilometers, rating:4.4, total reviewer:104)
Clontarf Castle Hotel (distance: 9.26724243435518 kilometers, rating:4.5, total reviewer:5073)
Sandycove Castle (distance: 11.6217343658077 kilometers, rating:4.5, total reviewer:24)
Castle (distance: 6.943123269967969 kilometers, rating:4, total reviewer:2)
Wooden Castle (distance: 8.662582766435369 kilometers, rating:4.6, total reviewer:11)
Lambay Castle (distance: 11.901973302290772 kilometers, rating:4.3, total reviewer:27)
Sutton Castle (distance: 2.3206824179509207 kilometers, rating:4.2, total reviewer:9)
Archbold's Castle (distance: 12.507363732093697 kilometers, rating:4.3, total reviewer:8)

solution generator module's Output:

Based on the provided map and Google Maps context, The castle closest to Howth rail station is Howth Castle which is only 0.477 km away from Howth rail station.

Thus, the correct answer is A. Howth Castle

answer generator module's Output:

Based on the response from the solution generator, the answer should be Howth Castle, which seems logical and correct. The answer is **A. Howth Castle**.

Comments:

The Final answer A. Howth Castle is a correct Answer

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Listing 5 : MapAgent 回答 MapEval-Visual Nearby 查询的示例 绿色 : 正确答案。红色 : 错误答案。

上下文 :

问题 :

哪座城堡距离 Howth 火车站最近 ?

选项 : A.Howth Castle B . Mary Castle C . Claremont Castle D . Balscadden House

Planner agent 选择的模块 :

从模块目录中选择的模块及其执行顺序 :

1. visual_place_recognizer 模块

2. map_service 模块

3. solution_generator 模块

4. answer_generator 模块

Visual Place Recognizer 模块输出 :

图像的中心似乎是爱尔兰的 Howth。

Map Service 模块输出 :

执行 map_service 模块时, 它连接到内层 (Inner Layer) 中的 map-tool agent。

Map-Tool Agent 输出 :

选择 Nearby 工具以获取信息。

Nearby 工具的输出 :

从当前位置 Howth Market , 3 Harbour Rd , Howth , Dublin , D13 T8Y7 , Ireland 出发, 15 公里范围内有以下一些城堡距离 :

Howth Castle Estate (距离 : 0.4770576879845442 公里, 评分 : 4.1, 评论总数 : 714)

Malahide Castle & Gardens (距离 : 8.742973877916125 公里, 评分 : 4.6, 评论总数 : 15235)

Dublin Castle (距离 : 13.875697894024961 公里, 评分 : 4.3, 评论总数 : 33876)

Swords Castle (距离 : 12.548387008403804 公里, 评分 : 4.5, 评论总数 : 1764)

Dalkey Castle & Heritage Centre (距离 : 12.505523354933828 公里, 评分 : 4.6, 评论总数 : 911)

Monkstown Castle (距离 : 12.235452667377206 公里, 评分 : 4.4, 评论总数 : 104)

Clontarf Castle Hotel (距离 : 9.26724243435518 公里, 评分 : 4.5, 评论总数 : 5073)

Sandycove Castle (距离 : 11.6217343658077 公里, 评分 : 4.5, 评论总数 : 24)

Castle (距离 : 6.943123269967969 公里, 评分 : 4, 评论总数 : 2)

Wooden Castle (距离 : 8.662582766435369 公里, 评分 : 4.6, 评论总数 : 11)

Lambay Castle (距离 : 11.901973302290772 公里, 评分 : 4.3, 评论总数 : 27)

Sutton Castle (距离 : 2.3206824179509207 公里, 评分 : 4.2, 评论总数 : 9)

Archbold 's Castle (距离 : 12.507363732093697 公里, 评分 : 4.3, 评论总数 : 8)

solution generator 模块的输出 :

基于提供的地图和 Google Maps 上下文, 距离 Howth 火车站最近的城堡是 Howth Castle, 它距离 Howth 火车站仅 0.477 公里。

因此, 正确答案是 A . Howth Castle

answer generator 模块的输出 :

基于 solution generator 的响应, 答案应该是 Howth Castle, 这看起来合乎逻辑且正确。答案是 A. Howth Castle。

评论 :

最终答案 A . Howth Castle 是正确答案

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Listing 6: Example 1 how VLLM used API to guess correct answer **Green**: Correct Answer. **Red**: Wrong Answer.

Context:



Question:
How many churches are visible in the lower half of the map?

Options: A. Five B. Two C. Four **D. Three**

Response with using API(MapAgent):

Google Map Response:

There are some churches within 0.8 kilometers of the location Rindögatan 46, 113 57 Stockholm, Sweden, as shown below:
City Church (distance: 0.46526450497253263 kilometers, rating:4.6, total reviewer:94)
St. Peter's Church, Stockholm (distance: 0.584886996790922 kilometers, rating:4.5, total reviewer:42)
Adolf Fredriks församling (distance: 0.36249979086127565 kilometers, rating:4.5, total reviewer:341)

final answer from solution generator:

"Based on the map provided and the context from the google map module, the churches visible in the lower half of the map are:"
"1. St. Peter's Church, Stockholm "
"2. Adolf Fredriks Frsamling "
"3. City Church "
"Thus, the correct answer is: **(D) Three**"

Response without using API(Base Model):

Based on the provided map, there are four visible churches in the map. Thus, the answer is: **(C) Four**.

Observation:
The correct answer should be D. Three which api version can predict but the base model which do not have api not able to answer.

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Listing 6：示例 1，VLLM 如何使用 API 猜出正确答案。绿色：正确答案。红色：错误答案。

上下文：

问题：

地图下半部分可以看到多少座教堂？

选项：A. 五座 B. 两座 C. 四座 D. 三座

使用 API (MapAgent) 的响应：

Google Map 响应：

在瑞典斯德哥尔摩 113 57，Radmansgatan 46 这一位置 0.8 公里范围内有一些教堂，如下所示：

City Church (距离：0.46526450497253263 公里，评分：4.6，总评论数：94)

St. Peter's Church, Stockholm (距离：0.584886996790922 公里，评分：4.5，总评论数：42)

Adolf Fredriks Forsamling (距离：0.36249979086127565 公里，评分：4.5，总评论数：341)

来自解答生成器的最终答案：

“根据所提供的地图以及 google map 模块的上下文，地图下半部分可见的教堂是：”

“1. St. Peter's Church, Stockholm”

“2. Adolf Fredriks Frsamling”

“3. City Church”

“因此，正确答案是：(D) 三座**”

不使用 API (Base Model) 的响应：

根据所提供的地图，地图中有四座可见的教堂。因此，答案是：(C) 四座。

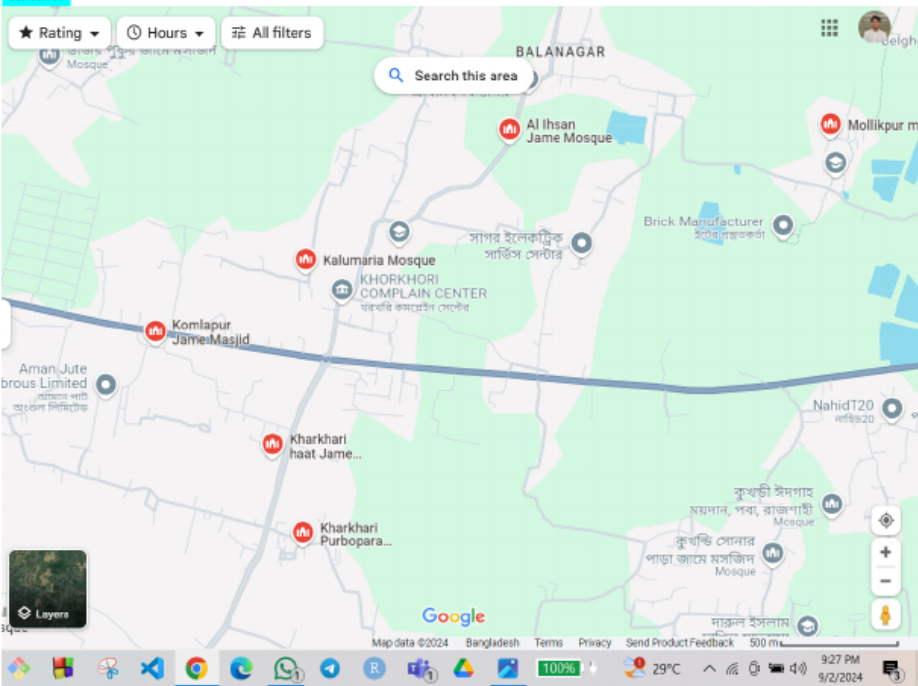
观察：

正确答案应该是 D. 三座，API 版本可以预测到，但没有 API 的 base model 无法回答。

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Listing 7: Example 2 how VLLM used API to guess correct answer **Green**: Correct Answer. **Red**: Wrong Answer.

Context:



Question:
I am at Sagor Electronix service center. What's the nearest mosque to me?

Options: A. Kalumaria Mosque **B. Al Ihsan Jame Mosque** C. Komlapur Jame Masjid D. Mollikpur mosque

Response with using API(MapAgent):

Google Map Response:
There are some mosque distance from the current location Sagor Electronix service center in below:
Al Ihsan Jame Mosque (distance: 0.5823095053737362 kilometers, rating:5, total reviewer:1)
Mollikpur mosque (distance: 1.1680710806133061 kilometers, rating:4.7, total reviewer:3)
Kalumaria Mosque (distance: 1.1879085293895582 kilometers, rating:4.3, total reviewer:4)
Lolithar Bazar Central Masjid (distance: 1.1566211819806247 kilometers, rating:0, total reviewer:0)
Komlapur Jame Masjid (distance: 1.8654544610349613 kilometers, rating:4.4, total reviewer:25)
Rajshahi University Central Mosque (distance: 4.459173025195116 kilometers, rating:4.9, total reviewer:221)
Maria Jama Masjid (distance: 1.8367913382567742 kilometers, rating:4, total reviewer:9)
Maherchandi East Jame Mosque (distance: 2.3601717780592995 kilometers, rating:5, total reviewer:5)
Vodra Central Jame Mosque (distance: 5.131809940553015 kilometers, rating:4.6, total reviewer:145)
Madina Nagar Jame Mosque (distance: 4.21188964565389 kilometers, rating:4.4, total reviewer:57)
Purbo Para Jame Mosque (distance: 3.614167437327442 kilometers, rating:4.2, total reviewer:26)
Meherchandi Purbapara Central Jame Mosque (distance: 3.4298745224188134 kilometers, rating:4.5, total reviewer:69)
Katakhal Bazar Jame Masjid (distance: 4.28092651862087 kilometers, rating:4.5, total reviewer:193)
Belghoria Jama Masjid (distance: 5.697832533309013 kilometers, rating:4.6, total reviewer:19)
Hazrapukur Jame Mosque (distance: 3.260713598861443 kilometers, rating:4.6, total reviewer:30)

final answer from solution generator:
The nearest mosque to your location, is ****Al Ihsan Jame Mosque**** with a distance of 0.582 kilometers.\n\nAnswer: ****B) Al Ihsan Jame Mosque****

Response without using API(Base Model):
Okay, I see your location on the map! The nearest mosque to Sagor Electronix service center appears to be **(C) Kalumaria Mosque.**

Observation:
The correct answer should be B. Three which api version can predict but the base model which do not have api not able to answer.

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Listing 7：示例 2，VLLM 如何使用 API 猜出正确答案。绿色：正确答案。红色：错误答案。

上下文：

问题：

我在 Sagor Electronix 服务中心。离我最近的清真寺是哪一个？

选项：A. Kalumaria Mosque B. Al Ihsan Jame Mosque C. Komlapur Jame Masjid D. Mollikpur mosque

使用 API (MapAgent) 的响应：

Google Map 响应：

从当前位置 Sagor Electronix 服务中心出发，有一些清真寺的距离如下：

Al Ihsan Jame Mosque (距离：0.5823095053737362 千米，评分：5，总评论数：1)

Mollikpur mosque (距离：1.1680710806133061 千米，评分：4.7，总评论数：3)

Kalumaria Mosque (距离：1.1879085293895582 千米，评分：4.3，总评论数：4)

Lolithar Bazar Central Masjid (距离：1.1566211819806247 千米，评分：0，总评论数：0)

Komlapur Jame Masjid (距离：1.8654544610349613 千米，评分：4.4，总评论数：25)

Rajshahi University Central Mosque (距离：4.459173025195116 千米，评分：4.9，总评论数：221)

Maria Jama Masjid (距离：1.8367913382567742 千米，评分：4，总评论数：9)

Maherchandi East Jame Mosque (距离：2.3601717780592995 千米，评分：5，总评论数：5)

Vodra Central Jame Mosque (距离：5.131809940553015 千米，评分：4.6，总评论数：145)

Madina Nagar Jame Mosque (距离：4.21188964565389 千米，评分：4.4，总评论数：57)

Purbo Para Jame Mosque (距离：3.614167437327442 千米，评分：4.2，总评论数：26)

Meherchandi Purbapara Central Jame Mosque (距离：3.4298745224188134 千米，评分：4.5，总评论数：69)

Katakhali Bazar Jame Masjid (距离：4.28092651862087 千米，评分：4.5，总评论数：193)

Belghoria Jama Masjid (距离：5.697832533309013 千米，评分：4.6，总评论数：19)

Hazrapukur Jame Mosque (距离：3.260713598861443 千米，评分：4.6，总评论数：30)

来自解决方案生成器的最终答案：

离你位置最近的清真寺是 Al Ihsan Jame Mosque，距离为 0.582 千米。
答案：(B) Al Ihsan Jame Mosque

不使用 API (Base Model) 的响应：

好的，我看到你在地图上的位置了！离 Sagor Electronix 服务中心最近的清真寺似乎是

(C) Kalumaria Mosque。

观察：

正确答案应该是 B。三个使用 API 的版本都能预测出来，但没有 API 的 base model 无法回答。

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